A First Course in Abstract Algebra

Part 4: Rings and Fields

Detailed worked solutions

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Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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18 Rings and Fields

Exercise 18.1: A product modulo 24

Multiply the integer representatives first: $12 \cdot 16 = 192$. Since $192 = 8 \cdot 24$, its remainder modulo 24 is 0. Thus the product in $\mathbb{Z}_{24}$ is $\boxed{0}$.

Exercise 18.2: A product modulo 32

The integer product is $16 \cdot 3 = 48 = 32 + 16$. Reduction modulo 32 therefore gives $\boxed{16}$.

Exercise 18.3: A negative product modulo 15

Compute $11(-4) = -44$. Adding the multiple 45 of 15 gives $-44 \equiv 1 \pmod{15}$. The product is $\boxed{1}$.

Exercise 18.4: A negative product modulo 26

We have $20(-8) = -160 = -7 \cdot 26 + 22$. Therefore the representative in $\{0, \ldots, 25\}$ is $\boxed{22}$.

Exercise 18.5: Multiplication in a product ring

Operations in $\mathbb{Z}_5 \times \mathbb{Z}_9$ are performed separately in each coordinate. Hence

$$(2, 3)(3, 5) = (6 \bmod 5, 15 \bmod 9) = \boxed{(1, 6)}.$$

Exercise 18.6: Two modular coordinates

In $\mathbb{Z}_4 \times \mathbb{Z}_{11}$ the product is

$$(-3, 5)(2, -4) = (-6, -20) = (2, 2),$$

because $-6 \equiv 2 \pmod{4}$ and $-20 \equiv 2 \pmod{11}$.

Exercise 18.7: The multiples of an integer

For $n \geq 1$, sums, negatives, and products of multiples of n remain multiples of n: $na - nb = n(a - b)$ and $(na)(nb) = n(nab)$. The associative and distributive laws are inherited from $\mathbb{Z}$. Thus $n\mathbb{Z}$ is a commutative ring. If it has unity u, applying $u \cdot n = n$ in the integers gives $u = 1$. Consequently it has unity exactly when $n = 1$, and it is never a field: when $n = 1$, the nonzero integer 2 has no inverse in $\mathbb{Z}$. If $n = 0$ is allowed, $0\mathbb{Z} = \{0\}$ is the zero ring, whose identity equals zero and which is not a field.

Exercise 18.8: Positive integers

Both addition and multiplication are closed on $\mathbb{Z}^+$. Nevertheless addition is not a group operation: 0 is absent and no positive integer has its additive inverse in the set. Thus these operations do not form a ring.

Exercise 18.9: The product of two integer rings

Addition and multiplication are componentwise, so each ring axiom follows in each coordinate from its counterpart in $\mathbb{Z}$. The result is a commutative ring with unity $(1,1)$. It is not a field: $(1,0)$ is nonzero but cannot have an inverse, because its product with any (a,b) has second coordinate zero.

Exercise 18.10: An even-integer coordinate

The subset $2\mathbb{Z}\times\mathbb{Z}$ is closed under subtraction and multiplication, so it is a commutative subring of $\mathbb{Z}\times\mathbb{Z}$. If (u,v) were its unity, multiplying $(2,0)$ would require $2u=2$, hence $u=1$, contrary to $u\in 2\mathbb{Z}$. It has no unity and is not a field.

Exercise 18.11: Integral coefficients with a square root

For integers a,b,c,d,

$$(a+b\sqrt{2})-(c+d\sqrt{2})=(a-c)+(b-d)\sqrt{2},\quad (a+b\sqrt{2})(c+d\sqrt{2})=(ac+2bd)+(ad+bc)\sqrt{2}.$$

Thus $\mathbb{Z}[\sqrt{2}]$ is a commutative subring of $\mathbb{R}$, with unity 1. It is not a field: an inverse of 2 would be $1/2$, which is not $a+b\sqrt{2}$ with integers a,b. Indeed $b\neq 0$ would make $\sqrt{2}$ rational, and $b=0$ would require $a=1/2$.

Exercise 18.12: Rational coefficients with a square root

The subtraction and multiplication formulas in Exercise 18.11 also hold for rational coefficients, proving that this set is a commutative ring with unity. For $a+b\sqrt{2}\neq 0$,

$$(a+b\sqrt{2})^{-1}=\frac{a-b\sqrt{2}}{a^2-2b^2}.$$

The denominator is nonzero: otherwise either $a=b=0$ or $\sqrt{2}=a/b$ would be rational. The displayed inverse again has rational coefficients. Therefore $\mathbb{Q}(\sqrt{2})$ is a field.

Exercise 18.13: Pure imaginary numbers

Addition is closed, but multiplication is not: $i\cdot i=-1$ is not purely imaginary. Since a ring requires a binary multiplication on its underlying set, this is not a ring under the indicated operations.

Exercise 18.14: Units in the integers

A unit $u\in\mathbb{Z}$ must satisfy $uv=1$ for an integer v. The only integer factors of 1 are 1 and -1. Thus the units are $\boxed{\{1,-1\}}$, each its own inverse.

Exercise 18.15: Units in a product of integer rings

The equation $(a,b)(c,d)=(1,1)$ is equivalent to $ac=1$ and $bd=1$. Therefore each of a,b must be 1 or -1, independently. The four units are $\boxed{(1,1),(1,-1),(-1,1),(-1,-1)}$.

Exercise 18.16: Units modulo five

Since 5 is prime, every nonzero residue is a unit. Explicitly, $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, and $4^{-1} = 4$, as multiplication verifies modulo 5. The units are $\boxed{1, 2, 3, 4}$.

Exercise 18.17: Units in the rational field

Every nonzero rational a/b has rational inverse b/a. Zero cannot multiply any rational to give 1. Thus the unit group is $\boxed{\mathbb{Q} \setminus \{0\}}$.

Exercise 18.18: Units in a three-factor ring

An inverse must work in all three coordinates. The first and last integer coordinates must therefore be ± 1, and the rational coordinate must be nonzero. All units are

$$\boxed{\{(\varepsilon, q, \delta) : \varepsilon, \delta \in \{1, -1\},\ q \in \mathbb{Q} \setminus \{0\}\}}.$$

The inverse is $(\varepsilon, q^{-1}, \delta)$.

Exercise 18.19: Units modulo four

The invertible residues are precisely those relatively prime to 4, namely $1, 3$. Directly, $1 \cdot 1 = 1$ and $3 \cdot 3 = 9 \equiv 1 \pmod 4$, whereas products with 0 or 2 cannot be odd. Thus the units are $\boxed{1, 3}$.

Exercise 18.20: The matrix ring over two elements

(a) Each of the four entries has two choices, giving $2^4 = \boxed{16}$ matrices. **(b)** A matrix over the field $\mathbb{Z}_2$ is invertible exactly when its determinant is nonzero. Its first column can be any of the three nonzero vectors; its second must be outside the span of the first, leaving two choices. The six units are

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Every listed determinant is 1 in $\mathbb{Z}_2$, and the counting argument shows that none are missing.

Exercise 18.21: A homomorphism that does not preserve unity

Use $\phi : \mathbb{Z} \longrightarrow \mathbb{Z} \times \mathbb{Z}$, $\phi(n) = (n, 0)$. Both addition and multiplication are preserved componentwise. However, $\phi(1) = (1, 0)$ is neither the zero $(0, 0)$ nor the target unity $(1, 1)$. The book's ring-homomorphism convention does not require unity preservation.

Exercise 18.22: The determinant map

For $n \geq 2$, determinants preserve multiplication but not addition. Take $A = \operatorname{diag}(1, 0, 0, \ldots)$ and $B = \operatorname{diag}(0, 1, 1, \ldots)$. Both determinants are zero, whereas $A + B = I_n$ has determinant 1. Hence $\det(A + B) \neq \det A + \det B$, and the map is not a ring homomorphism. For the exceptional case $n = 1$, the determinant is simply the entry, so it is an isomorphism $M_1(\mathbb{R}) \to \mathbb{R}$.

Exercise 18.23: Homomorphisms from the integers to themselves

Additivity forces $\phi(n) = n\phi(1)$ for every integer n, including negative integers. Set $k = \phi(1)$. Multiplicativity applied to $1^2 = 1$ gives $k^2 = k$, so $k = 0$ or 1. Both choices work. Thus the only maps are the zero map and the identity map.

Exercise 18.24: Homomorphisms into a product

Write $\phi(1) = (u, v)$. Additivity gives $\phi(n) = (nu, nv)$, and multiplicativity requires $(u^2, v^2) = (u, v)$. In $\mathbb{Z}$, an idempotent is 0 or 1. Consequently there are four homomorphisms:

$$n \mapsto (0,0), \quad n \mapsto (n,0), \quad n \mapsto (0,n), \quad n \mapsto (n,n).$$

Each preserves the two ring operations by direct substitution.

Exercise 18.25: Homomorphisms out of a product

Let $e_1 = (1,0)$ and $e_2 = (0,1)$. Since they are idempotent, their images u, v in $\mathbb{Z}$ lie in $\{0,1\}$. Since $e_1 e_2 = 0$, we must have $uv = 0$. Additivity then gives $\phi(a,b) = au + bv$. The possible pairs (u,v) are $(0,0), (1,0), (0,1)$, yielding exactly the zero map and the two coordinate projections.

Exercise 18.26: Three source coordinates

The three coordinate idempotents must map to 0 or 1 in $\mathbb{Z}$, and the product of any two different images must be zero. Therefore at most one image is 1. There are four choices: all zero, or one of the three coordinate positions equal to 1. These give the zero map and the three projections, so the answer is $\boxed{4}$.

Exercise 18.27: A false zero-product argument for matrices

The factorization is valid because I_3 commutes with X. The invalid step is inferring that a factor must be zero: matrix rings have zero divisors. For example,

$$X = \operatorname{diag}(1, 1, -1)$$

satisfies $X^2 = I_3$ while $X \neq I_3$ and $X \neq -I_3$. Here both $X - I_3$ and $X + I_3$ are nonzero, although their product is zero.

Exercise 18.28: A quadratic congruence modulo fourteen

Factor the polynomial as $(x+3)(x-2)$. A product is zero modulo 14 exactly when it is zero modulo both 2 and 7. Modulo 2 the original polynomial is $x(x+1)$, zero for both residues. Modulo 7, the two roots are 2 and $-3 = 4$. Lifting these residues to $\mathbb{Z}_{14}$ gives

$$\boxed{x = 2, 4, 9, 11}.$$

For example, at $x = 4$, neither $x + 3 = 7$ nor $x - 2 = 2$ is zero modulo 14, but their product is. This is precisely the zero-divisor issue in Exercise 18.27.

Exercise 18.29: Correcting the definition of a field

The missing requirement is commutativity of multiplication. A field is a *commutative* ring with unity $1 \neq 0$ such that every nonzero element has a multiplicative inverse; equivalently its nonzero elements form an abelian group under multiplication. The printed definition, without commutativity, describes a division ring.

Exercise 18.30: Correcting the definition of a unit

Magnitude is irrelevant in a general ring. In a ring with unity 1, an element u is a unit if there exists v in the ring with $uv = vu = 1$. Both equations are required in a possibly noncommutative ring.

Exercise 18.31: A product of nonzero elements that vanishes

In $\mathbb{Z}_6$, the residues 2 and 3 are nonzero, but $2 \cdot 3 = 6 = 0$. This supplies the requested example of nonzero zero divisors.

Exercise 18.32: A subring with its own unity

In $R = \mathbb{Z} \times \mathbb{Z}$, let $S = \mathbb{Z} \times \{0\}$. It is closed under subtraction and multiplication, so it is a subring. Its unity is $(1, 0)$, which is nonzero and differs from the unity $(1, 1)$ of R. Indeed $(1, 0)(a, 0) = (a, 0) = (a, 0)(1, 0)$.

Exercise 18.33: True or false: ring and field axioms

(a) True: a field is a ring with additional properties.

(b) False: $2\mathbb{Z}$ has no multiplicative identity.

(c) False: $\mathbb{Z}_2$ has just one unit, 1, since $-1 = 1$ there.

(d) False: $\mathbb{Z}_5$ has four units.

(e) True: $\mathbb{Z} \subset \mathbb{Q}$ is a subring but not a subfield, because $2^{-1} = 1/2 \notin \mathbb{Z}$.

(f) False: distributivity is an essential ring axiom, linking the two operations.

(g) True: commutative multiplication is part of the definition of a field.

(h) True: this is the nonzero multiplicative group in the field definition.

(i) True: the additive group of a ring is required to be abelian.

(j) True: the additive-group axiom provides an additive inverse for every element.

Exercise 18.34: Pointwise multiplication of functions

The operations are pointwise: $(fg)(x) = f(x)g(x)$ and $(f + g)(x) = f(x) + g(x)$. For every point x,

$$((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x).$$

Thus multiplication is associative. Likewise

$$(f(g+h))(x) = f(x)(g(x)+h(x)) = f(x)g(x) + f(x)h(x) = (fg+fh)(x),$$

and $((f+g)h)(x) = (fh+gh)(x)$. Equality at every input means equality of functions, proving the required associative and both distributive axioms.

Exercise 18.35: Evaluation preserves multiplication

For the evaluation map $\phi_a(f) = f(a)$, the definition of pointwise product gives

$$\phi_a(fg) = (fg)(a) = f(a)g(a) = \phi_a(f)\phi_a(g).$$

This verifies the multiplicative requirement, without assuming that a general function map preserves products.

Exercise 18.36: Ring isomorphism is an equivalence relation

Reflexivity: the identity map preserves both operations and is bijective. **Symmetry:** if $\phi : R \to S$ is an isomorphism, write $u = \phi(a)$, $v = \phi(b)$. Then $\phi^{-1}(u+v) = a+b$ and $\phi^{-1}(uv) = ab$, so its inverse is also an isomorphism. **Transitivity:** if $\psi : S \to T$ is an isomorphism as well, then

$$(\psi\phi)(a+b) = \psi(\phi(a)+\phi(b)) = (\psi\phi)(a) + (\psi\phi)(b),$$

and the same calculation with products proves multiplicativity. The composition is bijective. These are exactly the three equivalence-relation properties.

Exercise 18.37: The group of units

Let u, v be units. Their product is a unit because

$$(uv)(v^{-1}u^{-1}) = 1 = (v^{-1}u^{-1})(uv).$$

Thus multiplication is closed on the units; its associativity is inherited from the ring. The unity is a unit, and the inverse of a unit is itself a unit, with inverse u. Consequently all group axioms hold. The order reversal in $(uv)^{-1}$ matters when multiplication is not commutative.

Exercise 18.38: A difference-of-squares criterion

If R is commutative, expansion gives $(a+b)(a-b) = a^2 - ab + ba - b^2 = a^2 - b^2$. Conversely, suppose the stated identity holds for every a, b. Expanding its right side and canceling a^2 and $-b^2$ in the additive group gives $-ab + ba = 0$, hence $ab = ba$. Therefore multiplication is commutative.

Exercise 18.39: Zero multiplication

Define $ab = 0$ for every pair. This is a closed multiplication, and both $(ab)c$ and $a(bc)$ are zero. Furthermore $a(b+c) = 0 = 0+0 = ab+ac$ and $(a+b)c = 0 = ac+bc$. Together with the assumed abelian additive group, these facts prove all ring axioms.

Exercise 18.40: Two pairs of nonisomorphic rings

The rings $2\mathbb{Z}$ and $3\mathbb{Z}$. An additive isomorphism must carry the additive generator 2 to either 3 or -3. If $\phi(2) = 3\varepsilon$ with $\varepsilon = \pm 1$, additivity gives $\phi(4) = 6\varepsilon$, whereas multiplicativity gives $\phi(4) = \phi(2)^2 = 9$. These cannot agree.

The fields $\mathbb{R}$ and $\mathbb{C}$. Any field isomorphism carries unity to unity: its image of 1 acts as an identity on the whole target. Therefore a supposed map $\phi : \mathbb{C} \to \mathbb{R}$ would give $\phi(i)^2 = \phi(-1) = -1$. No real number has square -1. Hence these fields cannot be isomorphic.

Exercise 18.41: The prime-characteristic binomial identity

In a commutative ring the binomial expansion is valid. For $1 \le k \le p-1$, the binomial coefficient $\binom{p}{k}$ is divisible by the prime p: its denominator $k!(p-k)!$ has no factor p, whereas its numerator $p!$ has one. In $\mathbb{Z}_p$ all these intermediate coefficients vanish. Thus

$$(a+b)^p = a^p + \sum_{k=1}^{p-1} \binom{p}{k} a^{p-k} b^k + b^p = a^p + b^p.$$

Exercise 18.42: The unity of a subfield

Let e be the unity of a subfield $K \subset F$. It is nonzero, and $e^2 = e$ in F. Multiply this equation by the inverse of e in the field F to obtain $e = 1_F$. Cancellation in a field is why a subfield cannot exhibit the different-unity phenomenon of Exercise 18.32.

Exercise 18.43: Uniqueness of a multiplicative inverse

If b and c are both two-sided inverses of a, then

$$b = b1 = b(ac) = (ba)c = 1c = c.$$

Associativity and the two inverse equations justify every step. Thus a unit has exactly one two-sided inverse.

Exercise 18.44: Idempotents and a product ring

(a) If $a^2 = a$, $b^2 = b$, and multiplication is commutative, then $(ab)^2 = a^2 b^2 = ab$. Hence the product is idempotent.

(b) An idempotent residue x satisfies $x(x-1) \equiv 0$. In $\mathbb{Z}_6$, testing the six residues gives $\{0, 1, 3, 4\}$. In $\mathbb{Z}_{12}$, the requirement is that $x \equiv 0$ or 1 modulo each of 4 and 3, giving $\{0, 1, 4, 9\}$. Idempotence is coordinatewise, so all sixteen idempotents are

$$\boxed{\{0,1,3,4\} \times \{0,1,4,9\}}.$$

Exercise 18.45: An idempotent projection matrix

Put $B = (A^TA)^{-1}$, so $P = ABA^T$. Associativity of matrix multiplication, with no illicit reordering, gives

$$P^2 = ABA^TABA^T = AB(A^TA)BA^T = ABI_nA^T = ABA^T = P.$$

The dimensions are consistent: A is $m \times n$, B is $n \times n$, and P is $m \times m$. Thus it is an idempotent in $M_m(\mathbb{R})$. The printed reference to $n \times n$ matrices is a dimension slip: the formula for P gives $m \times m$.

Exercise 18.46: Adding nilpotent elements

Choose positive integers r, s with $a^r = 0$ and $b^s = 0$. Because the ring is commutative, expand $(a+b)^{r+s-1}$. In each term $a^kb^{r+s-1-k}$, either $k \geq r$, or else $k \leq r-1$ and $r+s-1-k \geq s$. The term is zero in either case. The entire expansion vanishes, so $a+b$ is nilpotent.

Exercise 18.47: Testing nilpotence using squares

If no nonzero nilpotent exists, an element satisfying $x^2 = 0$ must be zero. Conversely, suppose every square-zero element is zero. If a were nonzero nilpotent, choose the least $n \geq 2$ with $a^n = 0$. Then $b = a^{n-1} \neq 0$ by minimality, but $b^2 = a^{2n-2} = 0$ because $2n-2 \geq n$. This contradicts the square-zero condition.

Exercise 18.48: The subring test

A subring necessarily contains zero and is closed under subtraction and multiplication. Conversely suppose the three listed conditions hold. Setting $a = 0$ shows $-b \in S$; then $a + b = a - (-b) \in S$. Thus S is an additive subgroup of the abelian group $(R, +)$. Closure supplies a multiplication on S, and its associativity and both distributive laws are inherited from R. Hence S is a ring with the induced operations. A shared multiplicative unity is not required by the book's subring convention.

Exercise 18.49: Intersections of subrings and subfields

(a) Every subring in the family contains 0. If a, b are in their intersection, then $a - b$ and ab lie in every member, hence in the intersection. Apply Exercise 18.48.

(b) For a nonempty family of subfields of F, their common unity is 1_F by Exercise 18.42. The intersection is a subring and contains this nonzero unity. If $a \neq 0$ belongs to it, its unique inverse in F lies in every subfield, hence in the intersection. It is therefore a field. With the convention that an empty intersection is the ambient object, the empty-family conclusions hold as well.

Exercise 18.50: A one-sided annihilator is a subring

For $I_a = \{x : ax = 0\}$, first $a0 = 0$, so $0 \in I_a$. If $x, y \in I_a$, distributivity gives $a(x-y) = ax - ay = 0$. Associativity gives $a(xy) = (ax)y = 0y = 0$. Thus the subring test applies. No commutativity of the ambient ring is needed.

Exercise 18.51: The subring generated by one element

Let H consist of all finite sums $\sum_{k=1}^{N} m_k a^k$ with integer coefficients. This is exactly the additive subgroup generated by the positive powers of a. It contains a and is closed under subtraction. Products remain in H, because distributivity expresses a product as a finite integer combination of powers a^{i+j}. Therefore H is a subring containing a, so $R_a \subseteq H$ by the definition of R_a as an intersection. Conversely every subring containing a contains all positive powers and their integer linear combinations, so $H \subseteq R_a$. The two sets are equal. A constant term is not included: subrings here need not contain the ambient unity.

Exercise 18.52: Two simultaneous congruences

For coprime r, s, the map $\mathbb{Z}_{rs} \to \mathbb{Z}_r \times \mathbb{Z}_s$, $[x]_{rs} \mapsto ([x]_r, [x]_s)$, is a ring isomorphism as in the text. In particular it is onto, so the pair $([m]_r, [n]_s)$ has a preimage. Any integer representative x of that preimage satisfies both required congruences. Concretely, choose integers u, v with $ur + vs = 1$; then $x = nur + mvs$ has residues m modulo r and n modulo s.

Exercise 18.53: The general Chinese remainder theorem

(a) For pairwise coprime positive moduli $b_1, \ldots, b_n$, put $B = \prod b_i$. The map

$$\Phi : \mathbb{Z}_B \longrightarrow \prod_{i=1}^{n} \mathbb{Z}_{b_i}, \qquad [x]_B \mapsto ([x]_{b_1}, \ldots, [x]_{b_n})$$

is a well-defined ring homomorphism. Its kernel is zero: if every b_i divides x, their pairwise coprimality implies $B \mid x$. Both finite sets have B elements, so injectivity implies bijectivity. This proves the direct-product isomorphism.

(b) Surjectivity applied to the tuple of residues a_i produces the desired integer solution. Adding a sufficiently large multiple of B makes it positive without changing any congruence. For an explicit formula, let $B_i = B/b_i$ and choose u_i with $u_i B_i \equiv 1 \pmod{b_i}$. Then $x = \sum_i a_i u_i B_i$ has residue a_j modulo b_j, because every other summand is divisible by b_j.

Exercise 18.54: Recovering commutative addition in a division ring

We must prove the missing additive commutativity. Distributivity and additive cancellation first show $a0 = 0 = 0a$, since, for example, $a0 = a(0 + 0) = a0 + a0$. The multiplicative identity 1 on the nonzero elements also satisfies $10 = 01 = 0$, so it acts as unity on all of S. Expand the same product in two ways:

$$(1 + 1)(a + b) = a + b + a + b = (1 + 1)a + (1 + 1)b = a + a + b + b.$$

Cancel the initial a and final b in the additive group to obtain $b + a = a + b$. Thus $(S, +)$ is abelian. Multiplication is associative when all factors are nonzero by the assumed group axiom; if any factor is zero both associated products are zero. The assumed distributive laws now give all ring axioms, with nonzero unity and inverses for every nonzero element. Hence S is a division ring.

Exercise 18.55: Boolean rings are commutative

For any a, idempotence of $-a$ gives $(-a)^2 = -a$. But $(-a)^2 = a^2 = a$, so $a = -a$, equivalently $2a = 0$. Next expand the identity $(a+b)^2 = a+b$:

$$a^2 + ab + ba + b^2 = a + b.$$

Since $a^2 = a$ and $b^2 = b$, cancellation yields $ab + ba = 0$. Therefore $ab = -ba = ba$, using the already proved fact that every element equals its negative. The ring is commutative.

Exercise 18.56: A Boolean ring of subsets

(a) Abbreviate $\varnothing, \{a\}, \{b\}, \{a,b\}$ by $0, A, B, S$. The addition is symmetric difference and multiplication is intersection:

$+$	0	A	B	S
0	0	A	B	S
A	A	0	S	B
B	B	S	0	A
S	S	B	A	0

$\cdot$	0	A	B	S
0	0	0	0	0
A	0	A	0	A
B	0	0	B	B
S	0	A	B	S

.

(b) Associate to each subset $T \subseteq S$ its indicator function $\chi_T : S \to \mathbb{Z}_2$. At each point, $\chi_{T \triangle U} = \chi_T + \chi_U$ and $\chi_{T \cap U} = \chi_T \chi_U$, with operations in $\mathbb{Z}_2$. The bijection between subsets and their indicator functions therefore transfers the pointwise ring axioms to these set operations. The additive identity is $\varnothing$, each subset is its own additive inverse, and the unity is S. Finally $T \cap T = T$, so every element is idempotent and this is a Boolean ring.

19 Integral Domains

Exercise 19.1: A cubic congruence

Factor $x^3 - 2x^2 - 3x = x(x-3)(x+1)$. Because $12 = 3 \cdot 4$, it vanishes modulo 12 exactly when it vanishes modulo both 3 and 4. Modulo 3 the admissible residues are $0, 2$; modulo 4 they are $0, 1, 3$. Pairing these conditions, or testing the twelve residues, gives

$$\boxed{x = 0, 3, 5, 8, 9, 11 \text{ in } \mathbb{Z}_{12}}.$$

For example $x = 8$ gives $8 \cdot 5 \cdot 9 = 360$, divisible by 12. Factoring alone does not permit setting one factor equal to zero in a ring with zero divisors.

Exercise 19.2: Solving linear equations in two prime fields

In $\mathbb{Z}_7$, $3^{-1} = 5$, since $3 \cdot 5 = 15 \equiv 1$; hence $x = 5 \cdot 2 = \boxed{3}$. In $\mathbb{Z}_{23}$, $3^{-1} = 8$, since $3 \cdot 8 = 24 \equiv 1$; hence $x = \boxed{16}$. Multiplication by a nonzero field element is bijective, so these solutions are unique.

Exercise 19.3: A quadratic modulo six

For $x = 0, 1, 2, 3, 4, 5$, the residues of $x^2 + 2x + 2$ modulo 6 are respectively $2, 5, 4, 5, 2, 1$. None is zero, so there are $\boxed{\text{no solutions}}$. This exhaustive check is valid because these are all six elements of the ring.

Exercise 19.4: A second quadratic modulo six

Evaluating $x^2 + 2x + 4$ at $x = 0, 1, 2, 3, 4, 5$ gives residues $4, 1, 0, 1, 4, 3$ modulo 6. Therefore the complete solution set is $\boxed{\{2\}}$.

Exercise 19.5: Characteristic of the even integers

The characteristic is 0. For every positive integer n, the sum of n copies of the element 2 is the nonzero integer $2n$. Thus no positive n annihilates every element.

Exercise 19.6: Characteristic of two integer coordinates

For every positive n, $n(1, 0) = (n, 0) \neq (0, 0)$. Hence there is no common positive additive annihilator, and $\boxed{\text{char}(\mathbb{Z} \times \mathbb{Z}) = 0}$.

Exercise 19.7: A finite and an infinite coordinate

Although the first factor $\mathbb{Z}_3$ has characteristic 3, the element $(0, 3)$ in $\mathbb{Z}_3 \times 3\mathbb{Z}$ has infinite additive order. Its n-fold sum is $(0, 3n) \neq 0$ for every positive n. The product therefore has characteristic $\boxed{0}$.

Exercise 19.8: Two characteristic-three factors

Three copies of any element of $\mathbb{Z}_3 \times \mathbb{Z}_3$ sum to zero. One or two copies of $(1,0)$ do not. The least common positive annihilator is consequently $\boxed{3}$.

Exercise 19.9: Relatively prime finite characteristics

A positive integer annihilates $\mathbb{Z}_3 \times \mathbb{Z}_4$ exactly when it is divisible by both 3 and 4. The least such integer is $\operatorname{lcm}(3,4) = \boxed{12}$.

Exercise 19.10: A least common multiple of characteristics

The characteristic of $\mathbb{Z}_6 \times \mathbb{Z}_{15}$ is $\operatorname{lcm}(6,15) = \boxed{30}$. Indeed $n(1,0) = 0$ requires $6 \mid n$, while $n(0,1) = 0$ requires $15 \mid n$; these two conditions are also sufficient to annihilate every element.

Exercise 19.11: A fourth power in characteristic four

Commutativity allows the binomial theorem:

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

In characteristic 4, coefficients are reduced modulo 4: $4 = 0$ and $6 = 2$. Thus the simplified expression is $\boxed{a^4 + 2a^2b^2 + b^4}$. The middle term need not vanish.

Exercise 19.12: A fifth power in characteristic three

The binomial coefficients for exponent 5 are $1, 5, 10, 10, 5, 1$, reducing modulo 3 to $1, 2, 1, 1, 2, 1$. Therefore

$$\boxed{(a+b)^5 = a^5 + 2a^4b + a^3b^2 + a^2b^3 + 2ab^4 + b^5}.$$

Each reduction is justified because $3r = 0$ for every element r of the ring.

Exercise 19.13: A sixth power in characteristic three

First $(a+b)^3 = a^3 + b^3$, since the intermediate binomial coefficients are 3. Squaring and using commutativity gives

$$\boxed{(a+b)^6 = (a^3+b^3)^2 = a^6 + 2a^3b^3 + b^6}.$$

This agrees with reducing the sixth-row binomial coefficients modulo 3.

Exercise 19.14: Exhibiting a matrix zero divisor

Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}.$$

Both matrices are nonzero, but $AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, as row-by-column multiplication verifies. Thus A is a divisor of zero in $M_2(\mathbb{Z})$.

Exercise 19.15: Correcting zero-divisor terminology

One must exclude zero factors. A nonzero element a is a divisor of zero if there is a nonzero element b with $ab = 0$ or $ba = 0$ (in a commutative ring these are the same condition). The mere equation $ab = 0$ does not suffice if, for instance, $b = 0$.

Exercise 19.16: Correcting characteristic

The characteristic is the *least positive integer* n such that $na = 0$ for every $a \in R$, if such a positive integer exists. If none exists, the characteristic is 0. The printed condition omits both minimality and the characteristic-zero case.

Exercise 19.17: True or false: integral domains

(a) False: $n\mathbb{Z}$, as a subring of $\mathbb{Z}$, never has nonzero zero divisors, regardless of primality.

(b) True: a field is commutative with nonzero unity, and $ab = 0$, $a \neq 0$ implies $b = a^{-1}0 = 0$.

(c) False: for positive n, $n\mathbb{Z}$ has characteristic 0, since $kn \neq 0$ for positive k.

(d) False: for $n > 1$, $n\mathbb{Z}$ has no unity, whereas $\mathbb{Z}$ has one; an onto isomorphism would transport unity. Equivalently compare the images of 1 and 1^2.

(e) True: an isomorphism transports a zero-product or a cancellation equation to the integral domain, where cancellation holds.

(f) True: the elements $1, 2 \cdot 1, 3 \cdot 1, \ldots$ are distinct in characteristic zero.

(g) False: in a product of two domains, $(1, 0)(0, 1) = 0$ with both factors nonzero.

(h) True: multiplying $ab = 0$ by the inverse of a would force $b = 0$.

(i) False in general: for $n > 1$, $n\mathbb{Z}$ lacks unity and is not an integral domain under the book's definition. For $n = 1$ it is a subdomain.

(j) False: $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but it lacks the inverse of 2 and is not a field.

Exercise 19.18: Examples in the six ring regions

The regions of Figure 19.10 have the following examples.

1. $\mathbb{Q}$: a field.

2. $\mathbb{Z}$: an integral domain that is not a field, because 2 is not a unit.

3. $\mathbb{Z}_6$: a commutative ring with unity but with $2 \cdot 3 = 0$.

4. $2\mathbb{Z}$: a commutative ring without unity.

5. $M_2(\mathbb{R})$: a ring with unity but not commutative, since $E_{12}E_{21} = E_{11} \neq E_{22} = E_{21}E_{12}$.

6. $M_2(2\mathbb{Z})$: noncommutative by the same calculation using $2E_{12}, 2E_{21}$, and without unity. A unity would have to act as identity on $2E_{11}$ and $2E_{22}$, forcing it to be I_2, whose entries are not all even.

Exercise 19.19: Five matrix criteria for being a zero divisor

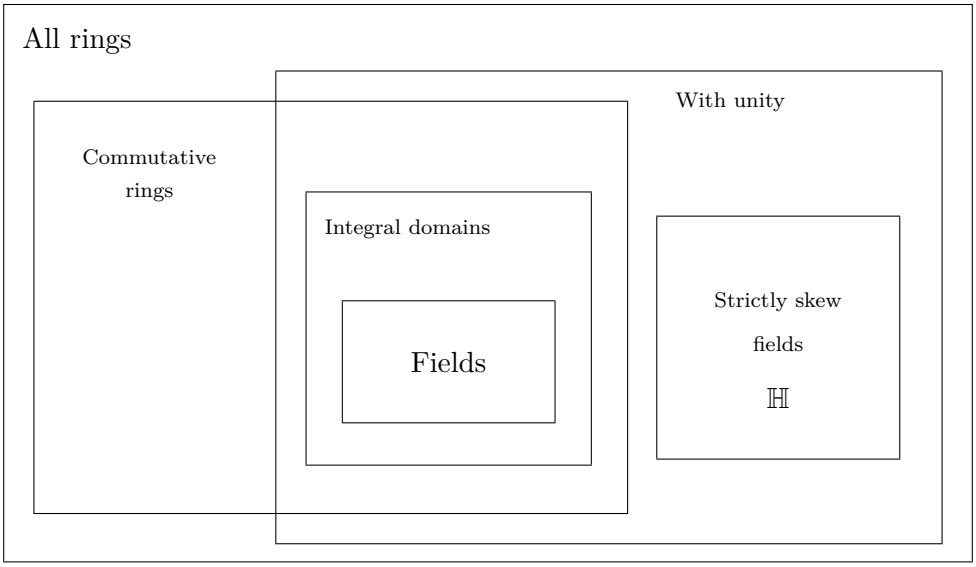
For a *nonzero* $A \in M_n(F)$, the following five equivalent tests characterize zero divisors:

$$\det A = 0; \quad \operatorname{rank} A < n; \quad \operatorname{Ker} A \neq \{0\}; \quad \text{columns of } A \text{ are dependent}; \quad A \text{ is not invertible.}$$

These are equivalent by the invertible-matrix theorem over a field. To connect them to the ring definition, if $Av = 0$ with $v \neq 0$, form B with first column v and all other columns zero. Then $B \neq 0$ and $AB = 0$. Conversely if $AB = 0$ with $B \neq 0$, some nonzero column v of B satisfies $Av = 0$. If instead $BA = 0$, invertibility would still be impossible. The exclusion $A \neq 0$ is essential because the book does not call zero itself a zero divisor.

Exercise 19.20: Locating strictly skew fields in the ring diagram

A strictly skew field is a division ring whose multiplication is not commutative. It therefore lies inside the rings-with-unity region and outside the commutative-rings region. Its set is disjoint from the integral domains and fields of the book's definitions. The quaternion division ring $\mathbb{H}$ supplies an example. The following inclusion diagram redraws the relevant regions using rectangles instead of circles.



Exercise 19.21: Synopsis of cancellation without zero divisors

If $ab = ac$ with $a \neq 0$, then $a(b - c) = 0$, so the absence of zero divisors forces $b - c = 0$ and hence $b = c$; the right-cancellation argument is identical with the factors reversed.

Exercise 19.22: Synopsis of the finite-domain theorem

For nonzero a in a finite integral domain, the map $x \mapsto ax$ is injective by cancellation and therefore surjective, so $ax = 1$ for some x, giving the required multiplicative inverse.

Exercise 19.23: Idempotents in a division ring

Both 0 and 1 are idempotent and are distinct. If $a^2 = a$ and $a \neq 0$, left multiplication by a^{-1} gives $a = 1$. Thus the only possibilities are exactly $\boxed{0, 1}$.

Exercise 19.24: Intersection of subdomains

A subdomain $D' \subseteq D$ must have the same unity as D: its nonzero unity e satisfies $e(e - 1_D) = 0$, hence $e = 1_D$ because D has no zero divisors. Thus an intersection of subdomains contains $1_D \neq 0$. It is a subring by the subring test, commutative by inheritance, and has no zero divisors because D has none. It is consequently a subdomain. For the empty family use the ambient domain as its intersection.

Exercise 19.25: A finite ring without zero divisors

Fix $a \neq 0$. The maps $x \mapsto ax$ and $x \mapsto xa$ are injective: equality of outputs gives a zero product after subtraction, so cancellation applies. Finiteness makes them onto. Hence there exist b, c with $ab = 1$ and $ca = 1$. They agree because $c = c(ab) = (ca)b = b$. Thus a has a two-sided inverse. This holds for every nonzero a, proving that R is a division ring; no commutativity was assumed.

Exercise 19.26: A unique generalized inverse forces a division ring

(a) Fix nonzero a and its unique b with $aba = a$. If either $ax = 0$ or $xa = 0$, then $axa = 0$, so $a(b + x)a = a$. Uniqueness implies $b + x = b$, hence $x = 0$. Thus there are no nonzero zero divisors.

(b) Associativity gives $a(bab)a = (aba)ba = aba = a$. Uniqueness of b gives $bab = b$.

(c) The element $e = ab$ is nonzero, since $aba = a \neq 0$, and $e^2 = (aba)b = ab = e$. For every x, $e(ex - x) = 0$ and $(xe - x)e = 0$. By (a), both differences are zero. Thus $ex = xe = x$, so e is a unity for the whole ring.

(d) Also ba is a nonzero idempotent and hence, by the same argument, is a unity. The uniqueness of unity gives $ba = ab = 1$. Every nonzero a therefore has a two-sided inverse, so R is a division ring.

Exercise 19.27: Characteristic of a subdomain

A subdomain $E \subseteq D$ has unity 1_D, as proved in Exercise 19.24. In any ring with unity the characteristic is the additive order of its unity, with infinite order recorded as characteristic zero. The sums $n1$ are identical whether computed in E or D, so their first positive zero, if any, occurs at the same n. Hence the two characteristics agree.

Exercise 19.28: The smallest subdomain

Let $P = \{n1_D : n \in \mathbb{Z}\}$. It contains 0 and 1_D, and

$$(m1_D) - (n1_D) = (m-n)1_D, \qquad (m1_D)(n1_D) = mn1_D.$$

It is therefore a commutative subring with nonzero unity, and has no zero divisors because it is contained in D. Thus it is a subdomain. Every subdomain contains the same unity 1_D and its integer sums and negatives, so every subdomain contains P.

Exercise 19.29: The characteristic of an integral domain

Suppose the positive characteristic is composite, say $n = rs$ with $1 < r, s < n$. Minimality gives $r1 \neq 0$ and $s1 \neq 0$, but

$$(r1)(s1) = rs1 = n1 = 0,$$

contradicting the absence of zero divisors. Also the characteristic cannot be 1, since $1 \neq 0$. Therefore the characteristic is either 0 or a prime.

Exercise 19.30: Adjoining a unity without changing characteristic

Write $K = \mathbb{Z}$ in characteristic zero, and $K = \mathbb{Z}_n$ in characteristic $n > 0$. On $S = R \times K$ use the stated operations. Integer scalar multiplication on R is well defined for residues modulo n because $nr = 0$.

(a) Addition is the direct-product abelian group. To check multiplication associativity, write three elements as $(r,k),(s,l),(t,m)$. The first coordinate of either associated product expands to

$$rst + k(st) + l(rt) + m(rs) + (kl)t + (km)s + (lm)r,$$

and the second coordinate is klm. The two distributive laws follow by expanding in the same way using distributivity in R and integer scalar multiplication. Thus this is a ring.

(b) Its unity is $(0,1)$, since

$$(0,1)(r,k) = (r,k) = (r,k)(0,1).$$

(c) If R has characteristic $n > 0$, both coordinates are killed by n, and the element $(0,1)$ has additive order exactly n. If R has characteristic zero, the integer coordinate of $(0,1)$ has infinite order. Therefore the characteristics agree. This includes the zero ring, with characteristic 1, if admitted.

(d) The map $r \mapsto (r,0)$ is injective and satisfies $(r,0)+(s,0) = (r+s,0)$ and $(r,0)(s,0) = (rs,0)$. It is consequently a ring isomorphism onto the subring $R \times \{0\}$.

20 Fermat's and Euler's Theorems

Exercise 20.1: A generator modulo 7

The nonzero residues form a multiplicative group of order 6. Successive powers of 3, with reduction modulo 7 after every multiplication, are

$$3^1, 3^2, \ldots, 3^6 \equiv 3, 2, 6, 4, 5, 1 \pmod 7.$$

Every nonzero residue appears once. Thus $\boxed{3}$ generates the entire multiplicative group.

Exercise 20.2: A generator modulo 11

The nonzero residues form a multiplicative group of order 10. Successive powers of 2, with reduction modulo 11 after every multiplication, are

$$2^1, 2^2, \ldots, 2^{10} \equiv 2, 4, 8, 5, 10, 9, 7, 3, 6, 1 \pmod{11}.$$

Every nonzero residue appears once. Thus $\boxed{2}$ generates the entire multiplicative group.

Exercise 20.3: A generator modulo 17

The nonzero residues form a multiplicative group of order 16. Successive powers of 3, with reduction modulo 17 after every multiplication, are

$$3^1, 3^2, \ldots, 3^{16} \equiv 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6, 1 \pmod{17}.$$

Every nonzero residue appears once. Thus $\boxed{3}$ generates the entire multiplicative group.

Exercise 20.4: A large power modulo twenty-three

Fermat's theorem applies because 23 is prime and does not divide 3. It gives $3^{22} \equiv 1 \pmod{23}$. Since $47 = 2 \cdot 22 + 3$,

$$3^{47} \equiv 3^3 = 27 \equiv \boxed{4} \pmod{23}.$$

Exercise 20.5: A large power modulo seven

Reduce the base: $37 \equiv 2 \pmod 7$. Fermat gives $2^6 \equiv 1 \pmod 7$, and $49 = 8 \cdot 6 + 1$. Therefore $37^{49} \equiv 2^{49} \equiv \boxed{2} \pmod 7$.

Exercise 20.6: A power whose exponent is itself a power

Since 19 is prime, $2^{18} \equiv 1 \pmod{19}$, so the outer exponent may be reduced modulo 18. The powers 2^k modulo 18 for $k = 1, \ldots, 6$ are $2, 4, 8, 16, 14, 10$, and $2^{k+6} \equiv 2^k$ for $k \geq 1$, because $2(2^6 - 1) = 126$ is divisible by 18. Thus $2^{17} \equiv 2^5 \equiv 14 \pmod{18}$. Now

$$2^{14} = 2^{10}2^4 \equiv 17 \cdot 16 = 272 \equiv 6 \pmod{19},$$

where $2^{10} = 1024 \equiv 17$. Consequently $2^{(2^{17})} + 1 \equiv \boxed{7} \pmod{19}$.

Exercise 20.7: Euler phi values through thirty

The value $\varphi(n)$ counts integers from 1 through n relatively prime to n; in particular $\varphi(1) = 1$. For computation, if the distinct prime divisors of n are p, use $\varphi(n) = n\prod_{p\mid n}(1 - 1/p)$. The complete table is

n	1	2	3	4	5	6	7	8	9	10
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4
n	11	12	13	14	15	16	17	18	19	20
$\varphi(n)$	10	4	12	6	8	8	16	6	18	8
n	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	12	10	22	8	20	12	18	12	28	8

Exercise 20.8: Euler phi for a prime square

Among $1, \ldots, p^2$, the integers not relatively prime to p^2 are exactly the p multiples $p, 2p, \ldots, p^2$. Removing them leaves $\boxed{\varphi(p^2) = p^2 - p = p(p-1)}$.

Exercise 20.9: Euler phi for a product of primes

If $p \neq q$, count the integers up to pq divisible by neither prime. There are q multiples of p, p multiples of q, and one multiple of both; inclusion–exclusion gives

$$\boxed{\varphi(pq) = pq - p - q + 1 = (p-1)(q-1)}.$$

If the primes are equal, the answer instead is $\varphi(p^2) = p(p-1)$ from Exercise 20.8. Distinctness cannot be silently assumed.

Exercise 20.10: Euler reduction modulo twenty-four

The base 7 is relatively prime to 24, and $\varphi(24) = 8$. Euler's theorem gives $7^8 \equiv 1 \pmod{24}$. Since $1000 = 125 \cdot 8$, the required remainder is $\boxed{1}$. One can also check directly that $7^2 = 49 \equiv 1 \pmod{24}$.

Exercise 20.11: A congruence with two residue solutions

The gcd is $\gcd(2, 4) = 2$, which divides 6. Divide the congruence by 2, including the modulus: $x \equiv 3 \pmod{2}$. Thus all integer solutions are $x = 1 + 2k$ with $k \in \mathbb{Z}$; the two incongruent solutions modulo 4 are $\boxed{1, 3}$.

Exercise 20.12: An invertible coefficient modulo fifteen

Reduce 22 modulo 15 to get $7x \equiv 5$. Since $7^{-1} = 13$ modulo 15, multiplication gives $x \equiv 65 \equiv 5 \pmod{15}$. Hence $\boxed{x = 5 + 15k, k \in \mathbb{Z}}$, one residue solution modulo 15.

Exercise 20.13: A congruence failing the gcd condition

Here $d = \gcd(36, 24) = 12$, but 12 does not divide 15. Every multiple $36x$ is divisible by 12, whereas $15 + 24k$ never is. Thus there are $\boxed{\text{no solutions}}$.

Exercise 20.14: Three residue solutions modulo twenty-four

Since $\gcd(45, 24) = 3$ divides 15, divide by 3 to obtain $15x \equiv 5 \pmod 8$, or $7x \equiv 5$. The inverse of 7 modulo 8 is 7, so $x \equiv 35 \equiv 3 \pmod 8$. Thus all integer solutions are $x = 3 + 8k$, and modulo 24 they are $\boxed{3, 11, 19}$.

Exercise 20.15: A second failure of the gcd condition

The coefficient 39 and modulus 9 have gcd 3. Since $3 \nmid 125$, the congruence has $\boxed{\text{no solutions}}$. Equivalently, its left side is zero modulo 3 but its right side is 2.

Exercise 20.16: One solution modulo nine

Reduce to $5x \equiv 8 \pmod 9$. Since $5 \cdot 2 \equiv 1$, multiply by 2: $x \equiv 16 \equiv 7 \pmod 9$. Thus $\boxed{x = 7 + 9k, k \in \mathbb{Z}}$.

Exercise 20.17: Five solutions modulo sixty-five

Here $\gcd(155, 65) = 5$, which divides 75. Divide by 5 to obtain $31x \equiv 15 \pmod{13}$, hence $5x \equiv 2$. Since $5^{-1} = 8$ modulo 13, $x \equiv 16 \equiv 3 \pmod{13}$. All integer solutions are $x = 3 + 13k$; the five residues modulo 65 are $\boxed{3, 16, 29, 42, 55}$.

Exercise 20.18: Thirteen solutions modulo one hundred thirty

The gcd is $\gcd(39, 130) = 13$, and $13 \mid 52$. Divide to obtain $3x \equiv 4 \pmod{10}$. Multiplying by $3^{-1} = 7$ gives $x \equiv 28 \equiv 8 \pmod{10}$. Hence all integer solutions are $\boxed{x = 8 + 10k}$, with $k \in \mathbb{Z}$; representatives modulo 130 are $8, 18, \ldots, 128$, thirteen in total.

Exercise 20.19: A factorial just below Wilson's factorial

Wilson's theorem gives $(p-1)! \equiv -1 \pmod p$. Since $(p-1)! = (p-1)(p-2)! \equiv -(p-2)!$, we obtain $\boxed{(p-2)! \equiv 1 \pmod p}$. For $p \geq 3$ the remainder is therefore 1.

Exercise 20.20: A factorial modulo thirty-seven

Write $36! = 36 \cdot 35 \cdot 34!$. Wilson's theorem and reduction of the last factors give $(-1)(-2)34! \equiv -1 \pmod{37}$, so $2 \cdot 34! \equiv -1$. The inverse of 2 is 19 modulo 37, and $-19 \equiv \boxed{18}$.

Exercise 20.21: A factorial modulo fifty-three

Wilson gives $52! \equiv -1 \pmod{53}$. Since $52! = 52 \cdot 51 \cdot 50 \cdot 49!$, we have $(-1)(-2)(-3)49! \equiv -1$. Thus $6 \cdot 49! \equiv 1$. The inverse of 6 modulo 53 is 9, because $6 \cdot 9 = 54 \equiv 1$. The remainder is $\boxed{9}$.

Exercise 20.22: A factorial modulo twenty-nine

Wilson gives $28! \equiv -1 \pmod{29}$. The four final factors are $28, 27, 26, 25 \equiv -1, -2, -3, -4$, whose product is 24. Hence $24 \cdot 24! \equiv -1$. Since $24 \cdot 23 = 552 \equiv 1$. Therefore $24! \equiv -23 \equiv \boxed{6} \pmod{29}$.

Exercise 20.23: True or false: Euler and congruences

(a) False: if $p \mid a$, then $a^{p-1} \equiv 0$, not 1.

(b) True: this is Fermat's theorem with its necessary nonzero-residue hypothesis.

(c) True: $\varphi(n)$ counts a subset of n integers, including $\varphi(1) = 1$.

(d) False as stated for all positive n: at $n = 1$, $\varphi(1) = 1 > 0 = n - 1$. It is true for $n \geq 2$, since n itself is then not coprime to n.

(e) True for $n \geq 2$, the usual nonzero-unity modular rings: the units are exactly residues with gcd 1 with n. If one admits $\mathbb{Z}_1$, the literal wording "positive integers less than n" omits its sole residue, so that case requires a separate convention.

(f) True: if u, v are units, $(uv)^{-1} = v^{-1}u^{-1}$.

(g) False: if ab is a unit in a commutative ring, $a[b(ab)^{-1}] = 1$ and similarly b is a unit; thus a product of two nonunits cannot be a unit.

(h) True by the same argument: a unit product would force the nonunit factor to be invertible.

(i) False: $0x \equiv 1 \pmod{p}$ has no solution.

(j) True: divide by d to obtain one solution modulo m/d, which lifts to exactly d distinct residue classes modulo m.

Exercise 20.24: The four units modulo twelve

The units are $1, 5, 7, 11$. Their table is

$\cdot$	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

.

Every nonidentity element has order 2, so this is not the cyclic group of order 4. It is the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$, for example via $1 \mapsto (0,0)$, $5 \mapsto (1,0)$, $7 \mapsto (0,1)$, $11 \mapsto (1,1)$.

Exercise 20.25: Synopsis of Fermat's theorem

For nonzero $a \in \mathbb{Z}_p$, Lagrange's theorem in the multiplicative group of the $p-1$ nonzero residues gives $a^{p-1} = 1$, which is precisely the desired congruence.

Exercise 20.26: Synopsis of Euler's theorem

The unit represented by an integer coprime to n lies in the multiplicative unit group of $\mathbb{Z}_n$, whose order is $\varphi(n)$, so Lagrange's theorem gives $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Exercise 20.27: Self-inverse nonzero residues

A self-inverse element satisfies $x^2 = 1$, hence $(x-1)(x+1) = 0$. Since $\mathbb{Z}_p$ is a field, one factor must vanish. Therefore $x = 1$ or $x = -1 = p-1$, and both do square to 1. For $p = 2$, these names describe the same element; for odd p, they are distinct.

Exercise 20.28: Wilson's congruence

For odd prime p, pair each nonzero residue other than 1 and -1 with its distinct multiplicative inverse. Every pair contributes product 1 to $(p-1)!$. By Exercise 20.27 the only unpaired factors are 1 and -1, giving $(p-1)! \equiv -1 \pmod{p}$. For $p = 2$, $1! = 1 \equiv -1 \pmod{2}$ directly.

For completeness, if an integer $n > 1$ satisfies this congruence, every prime divisor q of n must equal n: otherwise $q \leq n-1$ divides $(n-1)!$ and n, contradicting $(n-1)! \equiv -1 \pmod{q}$. Thus the converse also follows.

Exercise 20.29: A universal divisor of a polynomial value

Each prime p in $\{2, 3, 7, 13, 19, 37\}$ has $p - 1 \mid 36$. If $p \mid n$, then $n^{37} - n \equiv 0 \pmod{p}$. Otherwise Fermat gives $n^{36} \equiv 1 \pmod{p}$, again yielding $n^{37} - n = n(n^{36} - 1) \equiv 0$. These primes are distinct, so their product divides the difference:

$$2 \cdot 3 \cdot 7 \cdot 13 \cdot 19 \cdot 37 = \boxed{383838}.$$

Exercise 20.30: A larger universal divisor

Also $5 - 1 = 4$ divides 36, so the same Fermat argument shows $5 \mid n^{37} - n$ for every integer n. Since 5 is coprime to 383838, the larger number

$$\boxed{5 \cdot 383838 = 1919190}$$

divides every such difference.

21 The Field of Quotients of an Integral Domain

Exercise 21.1: Quotients of Gaussian integers

The field is $\boxed{\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\}}$. To see that every quotient lies here, rationalize its denominator:

$$\frac{m + ni}{r + si} = \frac{(mr + ns) + (nr - ms)i}{r^2 + s^2}, \qquad r + si \neq 0.$$

Both resulting coefficients are rational. Conversely, if $a, b \in \mathbb{Q}$, choose a common nonzero integer denominator d with $da, db \in \mathbb{Z}$. Then $a + bi = ((da) + (db)i)/d$ is a quotient of Gaussian integers. These two inclusions establish the description.

Exercise 21.2: Quotients with an irrational square root

The answer is $\boxed{\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}}$. A quotient rationalizes as

$$\frac{m + n\sqrt{2}}{r + s\sqrt{2}} = \frac{(mr - 2ns) + (nr - ms)\sqrt{2}}{r^2 - 2s^2}.$$

The denominator is nonzero whenever $r + s\sqrt{2} \neq 0$: otherwise irrationality of $\sqrt{2}$ forces $r = s = 0$. Thus the quotient has rational coefficients. Conversely, clearing the two rational denominators expresses any $a + b\sqrt{2}$ as a quotient of two members of $\mathbb{Z}[\sqrt{2}]$.

Exercise 21.3: Correcting the field-of-quotients definition

The missing condition is that *every element of* F be a quotient of elements of the embedded copy of D. Precisely, a field of quotients of D is a field F with an injective ring map $i : D \to F$ such that every $x \in F$ is $i(a)i(b)^{-1}$ for some $a, b \in D$, $b \neq 0$. Merely making all nonzero elements of D invertible allows larger fields such as $\mathbb{R}$ for $D = \mathbb{Z}$, and does not characterize the field of quotients.

Exercise 21.4: True or false: fraction fields

(a) True: every rational number is a quotient of integers.

(b) False: an irrational real number cannot be a quotient of integers.

(c) True: every real r is the quotient $r/1$ of real numbers.

(d) False: a quotient of real numbers is real, so it cannot equal i.

(e) True: if D is already a field, all its fractions $a/b = ab^{-1}$ remain in D.

(f) True: nonzero denominators must have nonzero products, and cancellation is used to establish transitivity of the equivalence relation.

(g) False: zero is an element of D and has no multiplicative inverse in a field.

(h) True: the image of a nonzero b has inverse represented by the fraction $1/b$.

(i) True: in a fraction field of D, the fractions a/b with $a, b \in D'$ form a subfield isomorphic to the fraction field of D'. Equality is the same cross-product equality in both domains.

(j) True: uniqueness of the fraction field up to isomorphism applies to $\mathbb{Z}$, and $\mathbb{Q}$ is one such field.

Exercise 21.5: Proper subdomains with the same fraction field

Take $D' = \mathbb{Z}$ and $D = \mathbb{Z}[1/2] = \{m/2^k : m \in \mathbb{Z},\ k \geq 0\}$. These are integral domains and D' is proper because $1/2 \in D \setminus D'$. The fraction field of either is $\mathbb{Q}$: fractions of elements of D are rational, and every rational is already a quotient of integers. Thus $F' = \mathbb{Q}$ works for both.

Exercise 21.6: Associativity of fraction addition

Write $[a, b]$ for the class $[(a, b)]$, and let $b, d, f \neq 0$. The addition formula gives

$$([a, b] + [c, d]) + [e, f] = [(ad + bc)f + (bd)e, bdf],$$

whereas

$$[a, b] + ([c, d] + [e, f]) = [a(df) + b(cf + de), bdf].$$

The numerators are both $adf + bcf + bde$ by the distributive and associative laws in D. The resulting ordered pairs are identical, hence so are their classes. This proves Part 2 of Step 3.

Exercise 21.7: The additive identity in the fraction construction

Using the prescribed operations,

$$[a, b] + [0, 1] = [a \cdot 1 + b \cdot 0, b \cdot 1] = [a, b].$$

Commutativity of addition, established in Part 1, also gives $[0, 1] + [a, b] = [a, b]$. Thus $[0, 1]$ is an additive identity, proving Part 3.

Exercise 21.8: The additive inverse of a fraction

Since $b \neq 0$, its square is nonzero, and $[-a, b]$ is an admissible fraction. Then

$$[a, b] + [-a, b] = [ab + b(-a), b^2] = [0, b^2] = [0, 1].$$

The last equality uses the defining equivalence: $0 \cdot 1 = b^2 \cdot 0$. Commutative addition supplies the equation in the other order. Hence $[-a, b]$ is the additive inverse, proving Part 4.

Exercise 21.9: Associativity of fraction multiplication

For three fractions, the multiplication rule gives

$$([a, b][c, d])[e, f] = [(ac)e, (bd)f], \qquad [a, b]([c, d][e, f]) = [a(ce), b(df)].$$

Associativity in D makes both numerators and both denominators identical. Nonzero denominator products follow from the integral-domain hypothesis. Therefore the classes are equal, proving Part 5.

Exercise 21.10: Commutativity of fraction multiplication

The products are $[a,b][c,d] = [ac, bd]$ and $[c,d][a,b] = [ca, db]$. Both coordinates agree because D is commutative. Thus the products are equal, which proves Part 6.

Exercise 21.11: Distributivity in the fraction construction

Let $x = [a,b]$, $y = [c,d]$, $z = [e,f]$. The prescribed formulas give

$$x(y+z) = [a(cf+de), bdf], \qquad xy + xz = [acbf + abde, b^2df].$$

The latter numerator is $b\,a(cf+de)$, so the two fractions are equivalent: their cross products are both $a(cf+de)b^2df$. Thus $x(y+z) = xy + xz$. Multiplication has already been proved commutative in Part 6, so

$$(y+z)x = x(y+z) = xy + xz = yx + zx.$$

This proves both distributive laws, Part 7.

Exercise 21.12: Partial quotients and a unity

Use classes $[r,t]$ of pairs in $R \times T$, with $[r,t] = [s,u]$ when $ru = st$. Transitivity follows by canceling a denominator from T; cancellation is valid because its elements are nonzero and not zero divisors. Products of denominators remain in T. The ordinary fraction addition and multiplication formulas are well defined by cross multiplication, just as in the text. They give a commutative ring by the same coordinate expansions as Exercises 21.6–21.11.

To view R inside it, fix any $t \in T$ and set $i(r) = [rt, t]$. The class is independent of t, because $(rt)u = (ru)t$. The formulas show that i preserves sums and products. If $i(r) = i(s)$, then $(r-s)t^2 = 0$; two cancellations of the regular element t give $r = s$. Thus i is an embedding.

(a) The element $e = [t,t]$ is independent of $t \in T$. It is a unity, since $[t,t][r,u] = [tr, tu] = [r,u]$ by cross multiplication. No unity in R was needed.

(b) For $u \in T$ and $t \in T$, the element $i(u) = [ut, t]$ has inverse $[t, ut]$, whose denominator lies in T. Their product is $[ut^2, ut^2] = e$. Hence every embedded element of T is a unit.

Exercise 21.13: Adjoining unity using regular powers

For a *nonzero* element a that is not a zero divisor, let $T = \{a^n : n \geq 1\}$. It is nonempty and multiplicatively closed. None of its elements is zero or a zero divisor: an equation $a^n x = 0$ permits n successive cancellations of a and forces $x = 0$. Exercise 21.12 therefore embeds R in the commutative ring $Q(R,T)$ with unity.

The qualification $a \neq 0$ is necessary for this particular argument: under the book's convention zero itself is not called a zero divisor, but its powers cannot serve as denominators. If the printed hypothesis is read as allowing only $a = 0$, use the general unity-adjoining construction of Exercise 19.30 instead. That construction applies to every ring, whereas localization additionally makes the chosen regular denominators invertible.

Exercise 21.14: Partial quotients of the ring modulo four

The allowed denominators $1, 3$ are already units in $\mathbb{Z}_4$, with $3^{-1} = 3$. Send $[r, t]$ to $rt^{-1} \in \mathbb{Z}_4$. Cross multiplication shows this is well defined and injective, and every residue occurs as $[r, 1]$. It is a ring isomorphism. Thus the partial-quotient ring has $\boxed{4}$ elements.

Exercise 21.15: Inverting powers of two

Send a formal fraction $[m, 2^n]$ to the real rational $m/2^n$. Cross multiplication makes this a well-defined injective ring map. Its image is

$$\boxed{\mathbb{Z}[1/2] = \{m/2^n : m \in \mathbb{Z},\ n \geq 0\}}.$$

Allowing $n = 0$ does not enlarge the image: $m = (2m)/2$. These are the dyadic rationals; denominators may contain only the prime 2 after reduction.

Exercise 21.16: Quotients of multiples of three

The map $[3m, 6^n] \mapsto 3m/6^n$ identifies the required ring with a subring of $\mathbb{Q}$. Every such fraction has denominator involving only 2 and 3, so its image is contained in $\mathbb{Z}[1/6]$. Conversely any $k/6^j$, $j \geq 0$, equals $6k/6^{j+1}$; the numerator $6k$ lies in $3\mathbb{Z}$ and the denominator is an allowed positive power. Therefore the image is exactly

$$\boxed{\mathbb{Z}[1/6] = \{k/6^j : k \in \mathbb{Z},\ j \geq 0\}}.$$

Exercise 21.17: The failure when denominators are zero divisors

The cross-product relation need not be transitive. In $R = \mathbb{Z}_6$, with $T = \{1, 2, 4\}$, consider

$$(0, 1), \qquad (0, 2), \qquad (3, 1).$$

The first two are related because $0 \cdot 2 = 1 \cdot 0$. The last two are related because $0 \cdot 1 = 2 \cdot 3 = 0$ modulo 6. But the first and third are not related, because $0 \cdot 1 = 0 \neq 3 = 1 \cdot 3$. Thus Step 1 already fails to define an equivalence relation. The transitivity proof would have canceled 2, which is a zero divisor. More general localization can be defined by a different relation, but then elements annihilated by denominators collapse to zero and the original ring need not embed.

22 Rings of Polynomials

Exercise 22.1: Addition and multiplication modulo eight

Add like powers, then reduce coefficients modulo 8:

$$f + g = 2x^2 - 3 = \boxed{2x^2 + 5}.$$

For the product, distribute before reducing:

$$fg = 8x^3 - 26x^2 + 28x - 10 = \boxed{6x^2 + 4x + 6} \quad \text{in } \mathbb{Z}_8[x].$$

The degree drops because the leading coefficient 8 vanishes modulo 8.

Exercise 22.2: Addition and multiplication in characteristic two

Here $f = g = x+1$. Thus $f+g = 2x+2 = \boxed{0}$. Expanding the product gives $fg = x^2+2x+1 = \boxed{x^2+1}$, since $2 = 0$ in $\mathbb{Z}_2$.

Exercise 22.3: Polynomial arithmetic modulo six

Addition gives

$$f + g = 5x^2 + 5x + 7 = \boxed{5x^2 + 5x + 1}.$$

Distributive multiplication over the integers gives

$$fg = 6x^4 + 13x^3 + 24x^2 + 17x + 12.$$

Reducing each coefficient modulo 6 gives $\boxed{x^3 + 5x}$. The expected degree 4 disappears because $2 \cdot 3 = 0$ in the coefficient ring.

Exercise 22.4: Polynomial arithmetic modulo five

After collecting terms,

$$f + g = 3x^4 + 2x^3 + 4x^2 + 5x + 6 = \boxed{3x^4 + 2x^3 + 4x^2 + 1}.$$

The unreduced product is

$$6x^7 + 12x^6 + 9x^5 + 10x^4 + 16x^3 + 22x^2 + 16x + 8.$$

Thus in $\mathbb{Z}_5[x]$,

$$\boxed{fg = x^7 + 2x^6 + 4x^5 + x^3 + 2x^2 + x + 3}.$$

Exercise 22.5: Counting short binary polynomials

Every polynomial of degree at most 3, including zero, is uniquely $a_0 + a_1x + a_2x^2 + a_3x^3$. Each coefficient has two choices in $\mathbb{Z}_2$, independently. The count is $\boxed{2^4 = 16}$. Allowing leading coefficients zero includes all smaller degrees.

Exercise 22.6: Counting short polynomials modulo five

Write the polynomial uniquely as $a_0 + a_1x + a_2x^2$, allowing leading zeros. Each of the three coefficients has five choices. The answer is $\boxed{5^3 = 125}$, including the zero polynomial.

Exercise 22.7: Evaluation at a real point

Evaluation means substituting the indicated input. Therefore $\phi_2(x^2 + 3) = 2^2 + 3 = \boxed{7}$.

Exercise 22.8: Evaluation at the imaginary unit

Substitute i and use $i^2 = -1$, $i^3 = -i$:

$$\phi_i(2x^3 - x^2 + 3x + 2) = 2(-i) - (-1) + 3i + 2 = \boxed{3 + i}.$$

Exercise 22.9: Evaluation of a product modulo seven

Evaluation preserves products, so evaluate the two factors separately at 3. They give $3^4 + 2 \cdot 3 = 87 \equiv 3$ and $3^3 - 3 \cdot 3^2 + 3 = 3$. Their product is $9 \equiv \boxed{2} \pmod 7$.

Exercise 22.10: Three evaluated factors modulo seven

At $x = 5$ the factors reduce modulo 7 to

$$5^2 + 2 = 27 \equiv 6, \quad 4 \cdot 5^2 + 3 = 103 \equiv 5, \quad 5^7 + 3 \cdot 5^2 + 1 \equiv 5 + 5 + 1 \equiv 4.$$

Here $5^7 \equiv 5$ by Fermat. Multiply the residues: $6 \cdot 5 \cdot 4 = 120 \equiv \boxed{1} \pmod 7$.

Exercise 22.11: Large exponents evaluated modulo seven

Since $4 \neq 0$ in $\mathbb{Z}_7$, Fermat gives $4^6 = 1$. Reduce the exponents: $106 \equiv 4$, $99 \equiv 3$, $53 \equiv 5 \pmod 6$. The residues $4^3 = 1$, $4^4 = 4$, $4^5 = 2$ in $\mathbb{Z}_7$ give

$$\phi_4(3x^{106} + 5x^{99} + 2x^{53}) = 3 \cdot 4 + 5 \cdot 1 + 2 \cdot 2 = 21 = \boxed{0}.$$

Exercise 22.12: Zeros over two elements

There are only two inputs. At 0, $x^2 + 1 = 1$; at 1, it is $1 + 1 = 0$ in $\mathbb{Z}_2$. Thus the complete zero set is $\boxed{\{1\}}$.

Exercise 22.13: Zeros of a cubic over seven elements

Evaluating at $0, 1, 2, 3, 4, 5, 6$ gives, respectively, residues $2, 5, 0, 0, 4, 4, 6$. Hence the zeros are exactly $\boxed{2, 3}$. In particular, a cubic may have a repeated root rather than three distinct roots.

Exercise 22.14: Zeros of a fifth-degree polynomial over five elements

For each $a \in \mathbb{Z}_5$, Fermat gives $a^5 = a$, so the value becomes $3a^3 + a^2 + 3a$. At $a = 0, 1, 2, 3, 4$ the residues are $0, 2, 4, 4, 0$. Therefore the complete zero set is $\boxed{\{0, 4\}}$.

Exercise 22.15: Zeros of a product

Since $\mathbb{Z}_7$ is a field, $f(a)g(a) = 0$ if and only if at least one factor is zero. Evaluating $f = x^3 + 2x^2 + 5$ at $0, \ldots, 6$ gives $5, 1, 0, 1, 3, 5, 6$, so its only zero is 2. Factoring $g = 3x^2 + 2x = x(3x + 2)$ gives zeros 0 and 4, because $3 \cdot 4 + 2 = 14 = 0$. Taking the union, the product has zeros $\boxed{0, 2, 4}$.

Exercise 22.16: Evaluation with prime-field exponent reduction

For nonzero $3 \in \mathbb{Z}_5$, $3^4 = 1$. The exponents satisfy $231 \equiv 3$, $117 \equiv 1$, $53 \equiv 1 \pmod 4$. Hence

$$\phi_3(x^{231} + 3x^{117} - 2x^{53} + 1) = 3^3 + 3 \cdot 3 - 2 \cdot 3 + 1 = 31 \equiv \boxed{1} \pmod 5.$$

Exercise 22.17: All zeros of a large-degree polynomial

At $x = 0$ the polynomial is zero, since every exponent is positive. For $x \neq 0$, use $x^4 = 1$ in $\mathbb{Z}_5$ and reduce exponents modulo 4:

$$2x^{219} + 3x^{74} + 2x^{57} + 3x^{44} = 2x^3 + 3x^2 + 2x + 3.$$

This cubic factors as $(x^2 + 1)(2x + 3)$. Its zeros are $x = 2, 3$ from $x^2 = -1 = 4$, and $x = 1$ from $2x = -3 = 2$. Including the separately checked zero input, all zeros are $\boxed{0, 1, 2, 3}$. The exponent reduction was not applied blindly to $x = 0$.

Exercise 22.18: Correcting the definition of a polynomial

Add the condition that only finitely many coefficients are nonzero. A polynomial over R is a formal expression $\sum_{i \geq 0} a_i x^i$ with $a_i \in R$ and $a_i = 0$ for all sufficiently large i. Without that finiteness condition, the expression is a formal power series rather than necessarily a polynomial.

Exercise 22.19: The definition of a zero

The stated definition is correct: $a \in F$ is a zero of f when evaluation at a gives $\phi_a(f) = f(a) = 0$. More generally, if $F \subseteq E$, a zero in E is an element $a \in E$ satisfying the same equation under $\phi_a : F[x] \to E$. The domain of the evaluation map is the polynomial ring $F[x]$.

Exercise 22.20: Regrouping a polynomial in two variables

Collect the displayed expression by powers of x rather than powers of y. The x^4 terms contribute $(y+1)x^4$; the x^3 term is $3y^3x^3$; the x^2 terms give $(y^2-3)x^2$; and the linear terms give $(2y^3-6y^2-2y)x$. The remaining constant in x is $y^2 + 2$. Thus in $(\mathbb{Q}[y])[x]$ it is

$$\boxed{(y + 1)x^4 + 3y^3x^3 + (y^2 - 3)x^2 + (2y^3 - 6y^2 - 2y)x + (y^2 + 2)}.$$

Exercise 22.21: Six kernel elements for evaluation at five

The kernel consists of polynomials with rational coefficients vanishing at 5. Six distinct examples are

$$0, \quad x-5, \quad 2(x-5), \quad (x-5)^2, \quad x(x-5), \quad (x-5)^3.$$

Every displayed expression has value zero after substituting $x = 5$, which directly verifies membership. Their different coefficients or degrees show that the six polynomials are distinct.

Exercise 22.22: A positive-degree polynomial unit

In $\mathbb{Z}_4[x]$, take $u = 1 + 2x$. Its square is $u^2 = 1 + 4x + 4x^2 = 1$. Therefore u is a unit, with inverse itself, and has degree $1 > 0$. This phenomenon uses the zero-divisor coefficient 2.

Exercise 22.23: True or false: polynomial rings

(a) True: a formal polynomial is zero exactly when each coefficient is zero.

(b) True: commutative coefficient products allow the convolution formula for multiplication to be reversed.

(c) True: the product of nonzero leading coefficients in a domain remains nonzero, so nonzero polynomials cannot multiply to zero.

(d) True: regard a pair of nonzero zero-divisor coefficients as constant polynomials.

(e) False: multiplication cannot create a term of degree greater than $3 + 4 = 7$.

(f) False: in $\mathbb{Z}_4[x]$, $(2x^3)(2x^4) = 0$, so the degree need not be 7.

(g) True: evaluation is multiplicative, giving $(fg)(a) = f(a)g(a) = 0$.

(h) True: if $fg = 1$ over a field, degree additivity forces both degrees to be zero; the units are nonzero constants.

(i) True whenever x is interpreted as the polynomial indeterminate in a ring with unity: multiplication by x shifts each coefficient one place, so $xf = 0$ forces $f = 0$. The same coefficient-shift statement holds formally without a unity.

(j) False: $2x$ is a nonconstant zero divisor in $\mathbb{Z}_4[x]$, since $(2x)2 = 0$.

Exercise 22.24: A polynomial ring over a domain is a domain

The standard coefficient operations give a commutative ring $D[x]$ with unity the constant polynomial $1_D \neq 0$. Let f, g be nonzero, with leading terms $a_m x^m, b_n x^n$. The coefficient of x^{m+n} in fg is $a_m b_n$, which is nonzero because D has no zero divisors. Hence $fg \neq 0$. Thus $D[x]$ has no nonzero zero divisors and is an integral domain.

Exercise 22.25: Units in three polynomial rings

(a) If $fg = 1$ in $D[x]$, then $\deg f + \deg g = \deg 1 = 0$ by the leading-coefficient argument. Both degrees must be zero, so f is a constant whose value is a unit in D. Conversely every such constant has its constant inverse in $D[x]$.

(b) The units of $\mathbb{Z}$ are ± 1, so the units of $\mathbb{Z}[x]$ are precisely the two constant polynomials $1, -1$.

(c) Every nonzero element of the field $\mathbb{Z}_7$ is a unit. Thus the units of $\mathbb{Z}_7[x]$ are precisely the constant polynomials $1, 2, 3, 4, 5, 6$.

Exercise 22.26: Left distributivity from coefficient formulas

Write $f = \sum a_i x^i$, $g = \sum b_j x^j$, and $h = \sum c_j x^j$, extending coefficient lists by zeros. The coefficient of x^n in $f(g + h)$ is

$$\sum_{i+j=n} a_i(b_j + c_j) = \sum_{i+j=n} a_i b_j + \sum_{i+j=n} a_i c_j.$$

Distributivity in R gives the equality and its abelian addition allows regrouping the finite sum. The right side is exactly the coefficient of x^n in $fg + fh$. Equality for every coefficient proves $f(g + h) = fg + fh$.

Exercise 22.27: Formal differentiation as an additive map

(a) Coefficientwise differentiation gives $D(f+g) = Df+Dg$, so D is an additive-group homomorphism. It is not a ring homomorphism: $D(x^2) = 2x$ but $D(x)D(x) = 1$. These formal polynomials are distinct.

(b) If $Df = 0$, every coefficient ka_k with $k \geq 1$ vanishes. In characteristic zero the scalar k is nonzero and invertible in the field, so $a_k = 0$ for $k \geq 1$. Hence the kernel is exactly the constant polynomials F.

(c) For $g = \sum_{k=0}^{n} b_k x^k$, set $f = \sum_{k=0}^{n} (b_k/(k+1)) x^{k+1}$. All denominators are nonzero in the characteristic-zero field. Then $Df = g$, so the image is all of $F[x]$.

Exercise 22.28: Evaluation in several variables

(a) For $\boldsymbol{a} = (a_1, \ldots, a_n) \in E^n$, define

$$\phi_{\boldsymbol{a}}\left(\sum_{\boldsymbol{k}} c_{\boldsymbol{k}} x_1^{k_1} \cdots x_n^{k_n}\right) = \sum_{\boldsymbol{k}} c_{\boldsymbol{k}} a_1^{k_1} \cdots a_n^{k_n}.$$

All sums are finite. Distributing products in the commutative field E proves that this map preserves addition and multiplication. It fixes the coefficients in F and sends x_i to a_i.

(b) Substitution yields $(-3)^2 2^3 + 3(-3)^4 2 = 9 \cdot 8 + 3 \cdot 81 \cdot 2 = \boxed{558}$.

(c) A zero in E^n is a tuple $\boldsymbol{a}$ with $\phi_{\boldsymbol{a}}(f) = f(a_1, \ldots, a_n) = 0$. A tuple, rather than a single scalar, is required.

Exercise 22.29: Pointwise operations on all functions

The zero function is the additive identity, and $(-\phi)(r) = -\phi(r)$ provides additive inverses. Addition is commutative and associative because these identities hold at every value r in R. For multiplication,

$$((\phi\psi)\theta)(r) = (\phi(r)\psi(r))\theta(r) = \phi(r)(\psi(r)\theta(r)) = (\phi(\psi\theta))(r).$$

Likewise evaluating either distributive law at r gives the corresponding law in R, for example

$$(\phi(\psi + \theta))(r) = \phi(r)(\psi(r) + \theta(r)) = (\phi\psi + \phi\theta)(r).$$

Closure is automatic because the operations in R have values in R. Thus the function set is a ring. Its multiplication is pointwise multiplication, as stipulated, not composition.

Exercise 22.30: Polynomial functions form a smaller kind of ring

(a) The zero function is polynomial. If functions u, v are represented by polynomials f, g, then $u - v$ is represented by $f - g$, and uv by fg. The subring test proves that P_F is a subring of the full function ring.

(b) If F has q elements, there are only q^q functions $F \to F$, hence only finitely many polynomial functions. But the formal polynomials $1, x, x^2, \ldots$ are infinitely many distinct members of $F[x]$. A bijection, and thus a ring isomorphism, is impossible. Distinct formal polynomials can induce the same function; over $\mathbb{Z}_p$, for instance, x^p and x do so.

Exercise 22.31: All functions on a finite field are polynomial

(a) Each input can be assigned any output independently. There are $\boxed{2^2 = 4}$ functions $\mathbb{Z}_2 \to \mathbb{Z}_2$, and $\boxed{3^3 = 27}$ functions $\mathbb{Z}_3 \to \mathbb{Z}_3$.

(b) Recording the two, or three, function values is an additive-group isomorphism to a Cartesian power. Thus the two groups are $\boxed{\mathbb{Z}_2 \times \mathbb{Z}_2}$ and $\boxed{\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3}$ respectively. Addition in each coordinate is the field addition.

(c) List $F = \{a_1, \ldots, a_q\}$. Given any function $u : F \to F$, define

$$f(x) = \sum_{i=1}^{q} u(a_i) \prod_{j \neq i} \frac{x - a_j}{a_i - a_j}.$$

Every denominator is nonzero because the a_i are distinct. At $x = a_k$, all summands but the kth contain a zero factor; the kth product equals 1. Therefore $f(a_k) = u(a_k)$ for every k. This polynomial represents u, proving $F^F = P_F$.

23 Factorization of Polynomials over a Field

Exercise 23.1: Polynomial division 1

Work in $\mathbb{Z}_7[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 1. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
x^4	$x^5 + 3x^4 + 4x^2 + 4x + 2$
x^3	$x^4 + 3x^3 + 4x^2 + 4x + 2$
x^2	$x^3 + 4x + 2$
x	$5x^2 + 2$
5	$4x + 3$

Thus $\boxed{q(x) = x^4 + x^3 + x^2 + x + 5}$ and $\boxed{r(x) = 4x + 3}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.2: Polynomial division 2

Work in $\mathbb{Z}_7[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 5. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$5x^4$	$x^4 + 4x^2 + 4x + 2$
$5x^2$	$4x^3 + 5x^2 + 4x + 2$
$6x$	$x + 2$

Thus $\boxed{q(x) = 5x^4 + 5x^2 + 6x}$ and $\boxed{r(x) = x + 2}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.3: Polynomial division 3

Work in $\mathbb{Z}_{11}[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 6. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$6x^4$	$3x^4 + 3x + 6$
$7x^3$	$4x^3 + 3x + 6$
$2x^2$	$9x^2 + 3x + 6$
$10x$	$4x + 6$
2	4

Thus $\boxed{q(x) = 6x^4 + 7x^3 + 2x^2 + 10x + 2}$ and $\boxed{r(x) = 4}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.4: Polynomial division 4

Work in $\mathbb{Z}_{11}[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 9. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$9x^2$	$3x^3 + x^2$
$5x$	$6x^2 + x$
10	2

Thus $\boxed{q(x) = 9x^2 + 5x + 10}$ and $\boxed{r(x) = 2}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.5: All multiplicative generators modulo 5

The group has order 4. The element 2 has this full order: it satisfies $2^4 = 1$, and the checks $2^2 \equiv 4 \pmod 5$ exclude every proper divisor of the order (check exponents $(4)/q$ for each prime q dividing 4). In a cyclic group of order 4, the generators are exactly g^k with $\gcd(k, 4) = 1$. Here those exponents are $1, 3$, yielding the complete generator set

$$\boxed{\{2, 3\}}.$$

Exercise 23.6: All multiplicative generators modulo 7

The group has order 6. The element 3 has this full order: it satisfies $3^6 = 1$, and the checks $3^3 \equiv 6, 3^2 \equiv 2 \pmod 7$ exclude every proper divisor of the order (check exponents $(6)/q$ for each prime q dividing 6). In a cyclic group of order 6, the generators are exactly g^k with $\gcd(k, 6) = 1$. Here those exponents are $1, 5$, yielding the complete generator set

$$\boxed{\{3, 5\}}.$$

Exercise 23.7: All multiplicative generators modulo 17

The group has order 16. The element 3 has this full order: it satisfies $3^{16} = 1$, and the checks $3^8 \equiv 16 \pmod{17}$ exclude every proper divisor of the order (check exponents $(16)/q$ for each prime q dividing 16). In a cyclic group of order 16, the generators are exactly g^k with $\gcd(k, 16) = 1$. Here those exponents are $1, 3, 5, 7, 9, 11, 13, 15$, yielding the complete generator set

$$\boxed{\{3, 5, 6, 7, 10, 11, 12, 14\}}.$$

Exercise 23.8: All multiplicative generators modulo 23

The group has order 22. The element 5 has this full order: it satisfies $5^{22} = 1$, and the checks $5^{11} \equiv 22, 5^2 \equiv 2 \pmod{23}$ exclude every proper divisor of the order (check exponents $(22)/q$ for each prime q dividing 22). In a cyclic group of order 22, the generators are exactly g^k with $\gcd(k, 22) = 1$. Here those exponents are $1, 3, 5, 7, 9, 13, 15, 17, 19, 21$, yielding the complete generator set

$$\boxed{\{5, 7, 10, 11, 14, 15, 17, 19, 20, 21\}}.$$

Exercise 23.9: Four linear factors modulo five

Since $4 = -1$ in $\mathbb{Z}_5$,

$$x^4 + 4 = x^4 - 1 = (x^2 - 1)(x^2 + 1).$$

Now $x^2 - 1 = (x - 1)(x + 1)$ and $x^2 + 1 = x^2 - 4 = (x - 2)(x + 2)$. Therefore the requested linear factorization is

$$\boxed{(x - 1)(x + 1)(x - 2)(x + 2)}.$$

Exercise 23.10: Three linear factors modulo seven

Substitution shows that $x = -1$ is a zero. Division gives

$$x^3 + 2x^2 + 2x + 1 = (x + 1)(x^2 + x + 1).$$

In $\mathbb{Z}_7$, $(x - 2)(x + 3) = x^2 + x - 6 = x^2 + x + 1$. Thus

$$\boxed{x^3 + 2x^2 + 2x + 1 = (x + 1)(x - 2)(x + 3)}.$$

Expansion checks all coefficients modulo 7.

Exercise 23.11: A nonmonic cubic modulo eleven

The residues $3, 4, -3$ are zeros, suggesting the three factors. Their product expands as

$$2(x - 4)(x - 3)(x + 3) = 2(x - 4)(x^2 - 9) = 2x^3 - 8x^2 - 18x + 72.$$

Reducing modulo 11 gives $2x^3+3x^2-7x-5$, as required. Hence this is the complete linear factorization, including its leading coefficient 2.

Exercise 23.12: A reducible cubic modulo five

The value at $x = 2$ is $8 + 4 + 3 = 15 = 0$ modulo 5, so the cubic is not irreducible. Division by $x - 2$ gives $x^2 + 2x + 1 = (x + 1)^2$. Therefore

$$\boxed{x^3 + 2x + 3 = (x - 2)(x + 1)^2} \quad \text{in } \mathbb{Z}_5[x].$$

All three linear factors are irreducible, with a repeated factor explicitly recorded.

Exercise 23.13: An irreducible cubic modulo five

A cubic over a field is reducible exactly when it has a zero in the field. The values of $2x^3 + x^2 + 2x + 2$ at $0, 1, 2, 3, 4$ are $2, 2, 1, 1, 4$ in $\mathbb{Z}_5$, none zero. It is therefore irreducible. If a monic irreducible factor is desired, write it as

$$\boxed{2(x^3 + 3x^2 + x + 1)}.$$

The scalar 2 is a unit, and the cubic factor is irreducible; there is no factorization into positive lower degrees.

Exercise 23.14: A quadratic over three fields

The discriminant of $x^2 + 8x - 2$ is $8^2 - 4(1)(-2) = 72$. Its roots are $-4 \pm 3\sqrt{2}$, neither rational. A quadratic over a field is reducible if and only if it has a root there, so it is irreducible over $\mathbb{Q}$. Both roots are real, however, giving

$$x^2 + 8x - 2 = (x + 4 - 3\sqrt{2})(x + 4 + 3\sqrt{2}).$$

Thus it is reducible over $\mathbb{R}$ and consequently over $\mathbb{C}$.

Exercise 23.15: A quadratic with negative discriminant

For $x^2 + 6x + 12$, the discriminant is $36 - 48 = -12$. There is no real root, hence no rational root. It is irreducible over both $\mathbb{Q}$ and $\mathbb{R}$. Over $\mathbb{C}$, its roots are $-3 \pm i\sqrt{3}$, so

$$x^2 + 6x + 12 = (x + 3 - i\sqrt{3})(x + 3 + i\sqrt{3})$$

is a factorization and it is reducible there.

Exercise 23.16: A cubic over the rationals

A reducible cubic over $\mathbb{Q}$ would have a rational root. The rational-root theorem for the monic polynomial $x^3 + 3x^2 - 8$ limits candidates to $\pm 1, \pm 2, \pm 4, \pm 8$. Their values, in that order, are

$$-4, -6, 12, -4, 104, -24, 696, -328,$$

all nonzero. Thus there is no rational root and the cubic is irreducible over $\mathbb{Q}$.

Exercise 23.17: A quartic with no quadratic factor

The only possible rational roots of the monic integer polynomial are ± 1, and both give -20, so no linear factor exists. By Gauss's lemma, any remaining rational factorization could be taken as

$$(x^2 + ax + b)(x^2 + cx + d), \qquad a, b, c, d \in \mathbb{Z}.$$

Comparing the x^3 and constant coefficients gives $c = -a$ and $bd = 1$, so $(b, d) = (1, 1)$ or $(-1, -1)$. The x^2 coefficient is $b + d - a^2 = -22$. In the first case this forces $a^2 = 24$; in the second it forces $a^2 = 20$. Neither is an integer square. Thus there is no quadratic factorization either, and $x^4 - 22x^2 + 1$ is irreducible over $\mathbb{Q}$.

Exercise 23.18: An Eisenstein quadratic

For $x^2 - 12$, use the prime $p = 3$. It does not divide the leading coefficient 1, divides all other coefficients (including the zero linear coefficient), and $3^2 = 9$ does not divide -12. Eisenstein's criterion therefore applies and proves irreducibility over $\mathbb{Q}$.

Exercise 23.19: An Eisenstein cubic

Use $p = 3$: it does not divide the leading coefficient 8, it divides the other coefficients $6, -9, 24$, and 9 does not divide the constant coefficient 24. Thus all Eisenstein hypotheses hold and the cubic is irreducible over $\mathbb{Q}$.

Exercise 23.20: An Eisenstein polynomial of degree ten

Use $p = 3$ for $4x^{10} - 9x^3 + 24x - 18$. Although $3 \nmid 4$ and 3 divides all lower coefficients, $3^2 = 9$ divides the constant coefficient -18. This prime fails the square condition. No other prime divides all the lower nonzero coefficients, since their gcd is 3. Thus the polynomial satisfies no Eisenstein criterion in its displayed form. Failure of the criterion alone does not decide reducibility.

Exercise 23.21: Another Eisenstein polynomial of degree ten

For $2x^{10} - 25x^3 + 10x^2 - 30$, the prime 5 divides every lower coefficient but not the leading coefficient 2. Its square 25 does not divide -30. Thus Eisenstein applies with $p = 5$, proving irreducibility over $\mathbb{Q}$.

Exercise 23.22: All rational zeros of a quartic

The rational-root theorem permits only fractions with numerator dividing 10 and denominator dividing 6. Testing candidates finds $x = -5/2$ and $x = 2/3$. Exact division, or multiplication, gives

$$6x^4 + 17x^3 + 7x^2 + x - 10 = (2x + 5)(3x - 2)(x^2 + x + 1).$$

The final quadratic has discriminant $1 - 4 = -3$, so it has no rational zero. Consequently the complete rational zero set is $\boxed{\{-5/2,\ 2/3\}}$. The displayed factorization checks the candidate search and excludes further rational roots.

Exercise 23.23: Correcting irreducibility

A nonconstant polynomial $f \in F[x]$ is irreducible over F if it cannot be written as $f = gh$ with *both* $g, h \in F[x]$ of positive degree. The printed definition forbids even $f = 1 \cdot f$, which every polynomial has. Trivial factorizations by nonzero constant units must be allowed.

Exercise 23.24: An equivalent irreducibility definition

The stated definition is correct. A nonconstant f is nonzero. In any factorization $f = gh$, both factors are nonzero, so a factor belonging to F is a nonzero constant and hence a unit in $F[x]$. Requiring such a factor in every factorization is equivalent to forbidding a product of two positive-degree polynomials.

Exercise 23.25: True or false: irreducibility and root bounds

(a) True: a degree-one polynomial cannot be a product of two positive-degree polynomials.

(b) True for the same reason; the nonzero scalar factor 3 is a unit in $\mathbb{Q}[x]$.

(c) True: $\sqrt{3}$ is irrational, so this quadratic has no rational root.

(d) False: in $\mathbb{Z}_7$, $2^2 + 3 = 7 = 0$, so $x^2 + 3$ is reducible.

(e) True: degree additivity forces any polynomial unit over a field to be a nonzero constant.

(f) True: the source repeats the same assertion as in (e), so the same reasoning applies.

(g) True: a polynomial of degree n is nonzero and has at most n distinct zeros over a field.

(h) True: regard it as the same nonzero degree-n polynomial over the extension field E and apply the root bound there.

(i) True: $ax + b$, $a \neq 0$, has zero $-b/a \in F$.

(j) False: over an infinite field such as $\mathbb{Q}$, the zero polynomial vanishes at infinitely many inputs. The assertion becomes true if one requires the polynomial to be nonzero.

Exercise 23.26: Primes for which a specified linear factor divides

The factor theorem says that $x + 2 = x - (-2)$ divides the polynomial modulo p exactly when its value at -2 is zero modulo p. This value is

$$16 - 8 + 4 + 2 + 1 = 15.$$

Thus p must divide 15. The prime possibilities are exactly $\boxed{3, 5}$, and both do satisfy the condition.

Exercise 23.27: Irreducible quadratics over two elements

Every monic quadratic has form $x^2 + ax + b$. It has no root at 0 only if $b = 1$; then its value at 1 is a, which is nonzero only if $a = 1$. Thus the only irreducible quadratic is $\boxed{x^2 + x + 1}$. There is no further nonzero scalar multiple to list because 1 is the only nonzero scalar in $\mathbb{Z}_2$.

Exercise 23.28: Irreducible cubics over two elements

A cubic is irreducible exactly when it has no field root. Write it $x^3 + ax^2 + bx + c$. Avoiding root 0 requires $c = 1$; avoiding root 1 requires $a + b \neq 0$, so $(a, b) = (1, 0)$ or $(0, 1)$. The complete list is

$$\boxed{x^3 + x^2 + 1, \qquad x^3 + x + 1}.$$

Exercise 23.29: Irreducible quadratics over three elements

Test the monic quadratics $x^2 + ax + b$ at $0, 1, 2$. The monic ones with no zero are exactly

$$x^2 + 1, \quad x^2 + x + 2, \quad x^2 + 2x + 2.$$

Their respective value triples are $(1, 2, 2)$, $(2, 1, 2)$, and $(2, 2, 1)$. All quadratic leading coefficients are 1 or 2, and multiplying by a nonzero constant preserves irreducibility. Thus the full answer is these three polynomials and twice each of them, six in total.

Exercise 23.30: Irreducible cubics over three elements

A cubic is irreducible exactly when none of $0, 1, 2$ is a zero. The monic list, together with values at these inputs, is

$h(x)$	$h(0)$	$h(1)$	$h(2)$
$x^3 + 2x^2 + 1$	1	1	2
$x^3 + 2x^2 + x + 1$	1	2	1
$x^3 + 2x + 1$	1	1	1
$x^3 + x^2 + 2x + 1$	1	2	2
$x^3 + x^2 + 2$	2	1	2
$x^3 + x^2 + x + 2$	2	2	1
$x^3 + 2x + 2$	2	2	2
$x^3 + 2x^2 + 2x + 2$	2	1	1

.

To see completeness, write a monic cubic as $x^3 + ax^2 + bx + c$. Necessarily $c = 1$ or 2; testing the nine pairs (a, b) for each c and discarding zeros at 1 or 2 gives precisely these eight rows. The full list of irreducible cubics consists of these eight polynomials and their multiples by 2, sixteen in all.

Exercise 23.31: Counting irreducible quadratics over a prime field

There are p^2 monic quadratics. A reducible monic quadratic is uniquely $(x-a)(x-b)$ for an unordered pair of field elements, allowing $a = b$. There are $\binom{p}{2} + p = p(p+1)/2$ such pairs. Therefore the number of monic irreducible quadratics is

$$p^2 - \frac{p(p+1)}{2} = \frac{p(p-1)}{2}.$$

Each has $p - 1$ distinct nonzero scalar multiples, and every nonmonic quadratic is a unique scalar times a monic one. Thus the requested count of *all* irreducible quadratics is $\boxed{p(p-1)^2/2}$.

Exercise 23.32: Synopsis of the bound on roots

Each distinct zero supplies a distinct linear factor by the factor theorem, and repeatedly removing these factors lowers the degree by one each time; a nonzero degree-n polynomial cannot supply more than n such factors.

Exercise 23.33: Synopsis of cyclicity of finite field subgroups

By the finite abelian group theorem, write the finite multiplicative subgroup with invariant factors $d_1 \mid \cdots \mid d_r$. Every element then satisfies $x^{d_r} = 1$. The polynomial root bound gives $|G| \leq d_r$, while the product of invariant factors is $|G| \geq d_r$. Equality forces all earlier factors to be trivial, so the group is cyclic.

Exercise 23.34: A reducible prime-power polynomial

In $\mathbb{Z}_p$, Fermat gives $a^p = a$, and all intermediate binomial coefficients of exponent p vanish. Hence

$$(x + a)^p = x^p + a^p = x^p + a.$$

Since $p \geq 2$, this is a product of p linear factors, so it is not irreducible. This includes $a = 0$, when the polynomial is x^p.

Exercise 23.35: The reciprocal polynomial

Let $g(x) = a_n + a_{n-1}x + \cdots + a_0x^n$. Since $a \neq 0$, substitute a^{-1} and factor out a^{-n}:

$$g(a^{-1}) = a^{-n}(a_0 + a_1a + \cdots + a_na^n) = a^{-n}f(a) = 0.$$

Thus $1/a$ is a zero of the reciprocal polynomial, with the nonzero assumption justifying all inverse powers.

Exercise 23.36: The remainder theorem

Division by $x - a$ gives $f(x) = (x - a)q(x) + r(x)$, where $r = 0$ or $\deg r < 1$. Thus r is a constant. Evaluating at a yields $f(a) = 0 \cdot q(a) + r = r$. Hence the remainder is exactly the constant polynomial $\boxed{f(a)}$.

Exercise 23.37: Reduction of integer coefficients

(a) Reduce each coefficient modulo m. Sums are preserved coefficientwise; products are preserved because each coefficient of a product is a finite sum of coefficient products, and integer reduction preserves both operations. Every polynomial over $\mathbb{Z}_m$ can be lifted by choosing integer representatives for its finitely many coefficients, so the map is onto.

(b) Suppose instead that f is reducible over $\mathbb{Q}$. By Gauss's lemma it has a factorization $f = gh$ in $\mathbb{Z}[x]$ with $0 < \deg g, \deg h < n$. The leading coefficient of f remains nonzero modulo m by the degree hypothesis. Because it is the product of the leading coefficients of g, h, neither of those can become zero modulo m. Their reduced degrees are therefore unchanged. Reducing $f = gh$ then gives the forbidden factorization into two degrees below n, a contradiction. Thus f is irreducible over $\mathbb{Q}$.

(c) Modulo 5, the polynomial becomes $x^3 + 2x + 1$. Its values at $0, 1, 2, 3, 4$ are $1, 4, 3, 4, 3$, none zero. As a cubic over the field $\mathbb{Z}_5$, it is irreducible, and its degree remains 3. Part (b) proves that $x^3 + 17x + 36$ is irreducible over $\mathbb{Q}$.

24 Noncommutative Examples

Exercise 24.1: Addition in a three-element group algebra

Add the coefficients of each basis element, reducing in $\mathbb{Z}_5$:

$$(2e + 3a + 0b) + (4e + 2a + 3b) = 6e + 5a + 3b = \boxed{e + 0a + 3b}.$$

The group elements are formal basis labels; coefficients are combined only when their labels agree.

Exercise 24.2: Multiplication in a three-element group algebra

Since G is cyclic of order 3, its multiplication satisfies $a^2 = b$, $ab = ba = e$, and $b^2 = a$. Distribute in the given order:

$$(2e + 3a)(4e + 2a + 3b) = 8e + 4a + 6b + 12a + 6b + 9e.$$

Collecting coefficients gives $17e + 16a + 12b$. Reduction modulo 5 yields $\boxed{2e + a + 2b}$.

Exercise 24.3: A fourth power in the group algebra

Put $S = e + a + b$. For each fixed product label, exactly three of the nine terms in S^2 contribute to it, so $S^2 = 3S$. It follows that $S^4 = 9S^2 = 27S$. Therefore

$$(3S)^4 = 3^4 \cdot 27S = 2187S = 2S \quad \text{over } \mathbb{Z}_5.$$

The answer is $\boxed{2e + 2a + 2b}$.

Exercise 24.4: An ordered quaternion product

Use $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, and reverse these last three products with a minus sign. Then

$$(i + 3j)(4 + 2j - k) = 4i + 2k + j + 12j - 6 - 3i.$$

Collecting the four real coordinates gives $\boxed{-6 + i + 13j + 2k}$. Keeping the order of factors avoids the wrong signs in $ik = -j$ and $jk = i$.

Exercise 24.5: Quaternion powers and multiplication

Reduce the powers first: $i^2 = -1$, $j^3 = -j$, and $i^5 = i$. Thus

$$i^2 j^3 k j i^5 = (-1)(-j)kji = (jk)ji = iji = ki = \boxed{j}.$$

Each reassociation uses associativity, and no two noncommuting factors have been interchanged.

Exercise 24.6: The inverse of a pure quaternion

For $q = i + j$, its conjugate is $\overline{q} = -i - j$ and its squared norm is $1^2 + 1^2 = 2$. Hence

$$\boxed{(i+j)^{-1} = \frac{-i-j}{2}}.$$

For a direct check, $(i+j)^2 = i^2 + ij + ji + j^2 = -2$, so multiplication by the displayed inverse gives 1 on either side.

Exercise 24.7: The inverse of a quaternion product

Multiply the factors before inverting:

$$(1+3i)(4j+3k) = 4j + 3k + 12k + 9ik = -5j + 15k.$$

The squared norm is $25 + 225 = 250$, and the conjugate is $5j - 15k$. Therefore

$$\boxed{[(1+3i)(4j+3k)]^{-1} = \frac{j-3k}{50}}.$$

Indeed a pure quaternion $v = -5j + 15k$ has $v^2 = -250$, so $v(-v/250) = 1 = (-v/250)v$.

Exercise 24.8: A product in the group algebra of the symmetric group

Discard the zero coefficients. The factors are $A = \rho_1 + \mu_2 + \mu_3$ and $B = \rho_0 + \rho_1 + \mu_1 + \mu_3$. Using the source's right-to-left permutation composition, the required twelve products are

	ρ_0	ρ_1	μ_1	μ_3
ρ_1	ρ_1	ρ_2	μ_3	μ_2
μ_2	μ_2	μ_3	ρ_2	ρ_1
μ_3	μ_3	μ_1	ρ_1	ρ_0

.

The multiplicities of $\rho_0, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3$ are $1, 3, 2, 1, 2, 3$. Coefficients are in $\mathbb{Z}_2$, so even multiplicities disappear and odd ones become 1. Thus

$$\boxed{AB = \rho_0 + \rho_1 + 0\rho_2 + \mu_1 + 0\mu_2 + \mu_3}.$$

Exercise 24.9: The center of the nonzero quaternion group

Every nonzero real scalar commutes with every quaternion, so $\mathbb{R} \setminus \{0\}$ is contained in the center. Conversely let $q = a + bi + cj + dk$ be central. Comparing

$$qi = -b + ai + dj - ck, \qquad iq = -b + ai - dj + ck$$

gives $c = d = 0$. Then comparing $(a+bi)j = aj + bk$ and $j(a+bi) = aj - bk$ gives $b = 0$. Therefore $q = a$ is real and nonzero. The center is exactly $\boxed{\mathbb{R} \setminus \{0\}}$ under multiplication.

Exercise 24.10: Two other complex planes inside the quaternions

Take $F_j = \{a + bj : a, b \in \mathbb{R}\}$ and $F_k = \{a + bk : a, b \in \mathbb{R}\}$. Because $j^2 = k^2 = -1$, their induced addition and multiplication obey exactly the complex-number formulas. The maps $a + bi \mapsto a + bj$ and $a + bi \mapsto a + bk$ are bijective ring homomorphisms from $\mathbb{C}$, hence field isomorphisms. These two subsets differ from each other and from the usual $\{a + bi\}$: for example j belongs only to F_j among these three planes.

Exercise 24.11: True or false: noncommutative examples

(a) False: $E_{11}E_{22} = 0$ in $M_2(F)$, with both factors nonzero.

(b) False: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is nonzero but has no inverse.

(c) False as stated for every abelian group: for the trivial group, the identity endomorphism equals the zero endomorphism. For every nontrivial A, the identity map is a nonzero unity.

(d) False: $A = \mathbb{Z}$ has a nonzero identity endomorphism as unity.

(e) False: for nontrivial A, the zero endomorphism is not an automorphism, so these automorphisms do not even contain the additive identity of a subring.

(f) False: take $R = \mathbb{Z}_2$. The group ring of the infinite cyclic group then has characteristic 2, whereas $\mathbb{Z}$ has characteristic zero, so the rings are not isomorphic.

(g) True: both the scalar coefficients and group labels commute, so every pair of formal sums commutes on expansion.

(h) False: $ij = k$ but $ji = -k$, so multiplication is not commutative. They form a division ring.

(i) True: quaternion multiplication is associative, 1 is the identity, and $q^{-1} = \overline{q}/\|q\|^2$ exists for every nonzero q.

(j) False: the real scalars, and also the usual complex plane, are subrings that are fields.

Exercise 24.12: Two failures of commutative-field conclusions

(a) In $\mathbb{H}$, the degree-two polynomial $x^2 + 1$ vanishes at $i, -i, j, -j, k, -k$, already six distinct zeros. More generally every $bi + cj + dk$ with $b^2 + c^2 + d^2 = 1$ has square -1. Thus the commutative-field root bound fails in a strictly skew field.

(b) The eight quaternions $\{\pm 1, \pm i, \pm j, \pm k\}$ form a finite multiplicative subgroup. Every element has order at most 4, so none generates all eight elements. This quaternion group is not cyclic.

Exercise 24.13: The other zero-divisor equation for an endomorphism

The source map is $\phi(m, n) = (m + n, 0)$. Define $\theta(m, n) = (m, -m)$; this is a nonzero additive endomorphism of $\mathbb{Z} \times \mathbb{Z}$. Then

$$(\phi\theta)(m, n) = \phi(m, -m) = (0, 0)$$

for every (m, n). Thus $\phi\theta = 0$ with $\theta \neq 0$, giving the required zero product on the side opposite to the source example's $\psi\phi = 0$.

Exercise 24.14: Six units over any field

The six matrices

$$\begin{pmatrix}1&0\\0&1\end{pmatrix}, \quad \begin{pmatrix}1&1\\0&1\end{pmatrix}, \quad \begin{pmatrix}0&1\\1&0\end{pmatrix}, \quad \begin{pmatrix}0&1\\1&1\end{pmatrix}, \quad \begin{pmatrix}1&1\\1&0\end{pmatrix}, \quad \begin{pmatrix}1&0\\1&1\end{pmatrix}$$

are distinct because $0 \neq 1$ in a field. Their determinants are $1, 1, -1, -1, -1, 1$, respectively, all nonzero even in characteristic 2 (where $-1 = 1$). The invertible-matrix criterion proves that all six are units of $M_2(F)$.

Exercise 24.15: Endomorphisms of cyclic additive groups

An additive endomorphism of $\mathbb{Z}$ is uniquely determined by the integer $k = \phi(1)$, and then $\phi(n) = kn$ for every integer n. The map $k \mapsto \phi_k$ is a bijection $\mathbb{Z} \to \mathrm{End}(\mathbb{Z})$, with

$$\phi_k + \phi_l = \phi_{k+l}, \qquad \phi_k \circ \phi_l = \phi_{kl}.$$

Thus it is a ring isomorphism.

For $\mathbb{Z}_n$, the same argument uses $k = \phi([1]) \in \mathbb{Z}_n$ and $\phi([r]) = [r]k$. Every k defines a well-defined endomorphism, since $nk = 0$ automatically. Addition and composition again correspond to $k + l$ and kl modulo n, proving $\mathrm{End}(\mathbb{Z}_n) \cong \mathbb{Z}_n$.

Exercise 24.16: Endomorphisms of a two-dimensional binary group

View $\mathbb{Z}_2 \times \mathbb{Z}_2$ as a two-dimensional vector space over $\mathbb{Z}_2$. Every additive endomorphism is linear and is determined by arbitrary images of its two basis vectors. Each image has four choices, so there are $4^2 = 16$ endomorphisms. The componentwise product ring $\mathbb{Z}_2 \times \mathbb{Z}_2$ has only four elements. Different cardinalities rule out a ring isomorphism. Equivalently the endomorphism ring is $M_2(\mathbb{Z}_2)$, which is also noncommutative.

Exercise 24.17: The differentiation–multiplication commutator

Here $Xf = xf$ and $Yf = f'$ are endomorphisms of the additive polynomial group. For every polynomial f,

$$(YX - XY)f = (xf)' - xf' = f + xf' - xf' = f.$$

The product rule can be checked on each monomial x^n: its two terms are $(n+1)x^n$ and nx^n. Thus $YX - XY$ acts as the identity on every polynomial and equals the unity endomorphism $\boxed{1}$.

Exercise 24.18: The group ring of a one-element group

If $G = \{e\}$, every formal sum has the form re. The map $re \mapsto r$ is bijective. Furthermore $(re) + (se) = (r + s)e$ and $(re)(se) = (rs)e$, since $e^2 = e$. The map preserves both operations, proving $RG \cong R$.

Exercise 24.19: A matrix realization of the quaternions

Put

$$E = I_2, \qquad I = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The i in these matrix entries is the complex imaginary unit.

(a) Since quaternion $ij = k$, multiplicativity forces

$$\boxed{K = IJ = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}}.$$

(b) With $IJ = K$ used as this definition, the other eight multiplication-table equations to check are

$$I^2 = J^2 = K^2 = -E, \qquad JI = -K, \qquad JK = I, \qquad KJ = -I, \qquad KI = J, \qquad IK = -J.$$

Direct matrix multiplication verifies them: the three squares are $\mathrm{diag}(-1,-1)$, $JI = \mathrm{diag}(-i,i)$, $JK = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and the remaining reversed products have the stated signs. Identities involving E hold automatically. Real-linearity of the definition makes additivity automatic, and distributivity reduces multiplicativity for arbitrary quaternions to these basis products.

(c) We must also check injectivity, since surjectivity onto the image is automatic. The explicit matrix is

$$\phi(a + bi + cj + dk) = \begin{pmatrix} a + di & b + ci \\ -b + ci & a - di \end{pmatrix}.$$

If it is zero, real and imaginary parts of its entries give $a = d = b = c = 0$. Thus the kernel is zero, proving the required isomorphism onto $\phi[\mathbb{H}]$.

25 Ordered Rings and Fields

Exercise 25.1: The high-degree ordering of monomials

For P_{high}, the sign of a nonzero polynomial is the sign of its highest-degree coefficient. Thus $x - a$ is positive for every $a > 0$ in R, and $x^{n+1} - x^n$ is positive for every $n \geq 1$. The requested ordering is

$$\boxed{0 < a < x < x^2 < x^3 < \cdots < x^n < x^{n+1} < \cdots}.$$

The comparison $a < x$ concerns every positive constant, regardless of its size in the original ring.

Exercise 25.2: Laurent monomials in the low-degree ordering

The sign of a nonzero Laurent series is determined by the coefficient at its least exponent. For integers $i < j$, the difference $x^i - x^j$ has least term x^i with positive coefficient. Consequently $x^j < x^i$. All monomials are positive, and the full order is

$$\boxed{0 < \cdots < x^3 < x^2 < x < 1 < x^{-1} < x^{-2} < x^{-3} < \cdots}.$$

In particular x is positive and smaller than every positive constant of the coefficient field.

Exercise 25.3: Positive elements in the conjugate ordering

The ordering is transported by $\phi(m + n\sqrt{2}) = m - n\sqrt{2}$. Therefore an element is positive in this new ordering exactly when $m - n\sqrt{2} > 0$ in the usual real order. In integer-coefficient cases this means:

- if $n = 0$, then $m > 0$;
- if $n > 0$, then $m > 0$ and $m^2 > 2n^2$;
- if $n < 0$, then either $m \geq 0$, or $m < 0$ and $2n^2 > m^2$.

Squaring in these comparisons is valid only after the relevant sides have been made nonnegative. For example $-\sqrt{2}$ has $m = 0, n = -1$, so it is positive in this ordering.

Exercise 25.4: Ordering five polynomials, set 4

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{a < c < d < e < b}$. The deciding terms of its four successive differences are

$$5, 2x, 2, 3.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{d < b < a < e < c}$. The deciding terms of its four successive differences are

$$7x^2, 6x, 4x^2, 3x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.5: Ordering five polynomials, set 5

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{a < c < e < d < b}$. The deciding terms of its four successive differences are

$$1, 5x, 8x^2, 3x.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < b < a < d}$. The deciding terms of its four successive differences are

$$4x^5, 11x^4, 8x^3, x^5.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.6: Ordering five polynomials, set 6

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{c < a < b < e < d}$. The deciding terms of its four successive differences are

$$2, 3, 2x, 6x^3.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < a < b < d}$. The deciding terms of its four successive differences are

$$5x^5, 5x^2, x^3, 8x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.7: Ordering five polynomials, set 7

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{d < a < b < c < e}$. The deciding terms of its four successive differences are

$$3x, 2x^2, 2x, x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{d < c < e < a < b}$. The deciding terms of its four successive differences are

$$x^2, x^2, 5x^3, 4x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.8: Ordering five polynomials, set 8

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{e < a < c < b < d}$. The deciding terms of its four successive differences are

$$x, 3x^2, 2x^2, x.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{c < d < a < e < b}$. The deciding terms of its four successive differences are

$$x, 6x^3, x^2, 4x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.9: Ordering five polynomials, set 9

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{c < a < e < d < b}$. The deciding terms of its four successive differences are

$$x^3, 6x^2, 4x^3, 2.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < b < a < d}$. The deciding terms of its four successive differences are

$$8x^3, x^3, x, 4x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.10: Ordering Laurent monomials with coefficients

The least exponent controls the sign. The two negative terms come first: $-5x^{-2} < -3x^{-2}$. Among the positive terms, $4x < x^{-1} < 2x^{-1}$, because a positive x^{-1} term exceeds any positive multiple of x, and the coefficient 1 is less than 2 at exponent -1. Thus the increasing label order is $\boxed{b < d < e < a < c}$.

Exercise 25.11: Ordering five rational Laurent series

Use formal geometric-series division to expose each least term:

$$a = 1 + x + \cdots, \quad b = x^2 + \cdots, \quad c = x^{-1} + 1 + \cdots,$$
$$d = -x + \cdots, \qquad e = \tfrac{3}{4}x^{-1} - \tfrac{1}{2} + \cdots.$$

The negative series d comes first. The positive infinitesimal b is below a, whose constant term is 1. Both e and c have positive exponent--1 terms, so exceed a; their coefficients compare as $3/4 < 1$. Therefore $\boxed{d < b < a < e < c}$. The dots hide higher exponents, which cannot alter these deciding terms.

Exercise 25.12: Comparing equal constant terms in Laurent series

Expansion gives

$$a = \tfrac{5}{3}x^{-1} - \tfrac{26}{9} + \cdots, \qquad b = -\tfrac{1}{2} + \tfrac{5}{8}x + \cdots, \qquad c = \tfrac{7}{4} + \cdots,$$

$$d = \tfrac{9}{2} + \cdots, \qquad e = -\tfrac{1}{2} + \tfrac{2}{3}x + \cdots.$$

For b and e, the constant terms cancel; the coefficient of x in $e - b$ is $2/3 - 5/8 = 1/24 > 0$, proving $b < e$. Their negative constant terms put both below c and d. Since $7/4 < 9/2$, $c < d$, and the positive x^{-1} term puts a above them all. Thus $\boxed{b < e < c < d < a}$.

Exercise 25.13: A second comparison of equal constant terms

Formal expansions yield

$$a = 1 - 2x + \cdots, \quad b = 1 - \tfrac{10}{3}x + \cdots, \quad c = \tfrac{1}{4}x^{-1} - \tfrac{1}{16} + \cdots,$$

$$d = -\tfrac{1}{3}x^{-1} - \tfrac{1}{9} + \cdots, \qquad e = 4x + \cdots.$$

The negative x^{-1} term places d first, and the positive such term places c last. The positive infinitesimal e lies below the two series with constant term 1. Finally $a - b$ has least term $(4/3)x > 0$, so $b < a$. The increasing order is $\boxed{d < e < b < a < c}$.

Exercise 25.14: An ordered field containing nonreal numbers

Let $K = \mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R}$, with its induced real ordering, and let $L \subset \mathbb{C}$ be the isomorphic field in the question, containing $\sqrt[3]{2}(-1 + i\sqrt{3})/2$. If $\phi : K \to L$ is the stated isomorphism, declare $\phi(a) >_L 0$ exactly when $a > 0$ in K. The cone axioms transfer through the isomorphism, so this orders L, although L contains nonreal complex numbers. There is no contradiction: the order uses the algebraic field structure of L and is not a restriction of an order on all of $\mathbb{C}$. The obstruction $i^2 = -1$ rules out ordering $\mathbb{C}$ itself, not every nonreal subfield.

Exercise 25.15: True or false: orderings

(a) True: $1 = 1^2$ is positive; positive integer sums are then positive, and trichotomy forces the negatives to be negative.

(b) True: every usual positive real is a nonzero square, so must be positive in any field ordering. Trichotomy fixes all signs.

(c) False: $\mathbb{Q}(\sqrt{2})$ has its usual ordering and the ordering transported by conjugation $\sqrt{2} \mapsto -\sqrt{2}$, which disagree on $\sqrt{2}$.

(d) True: positive integers are positive, their inverses are positive, and hence all usual positive rationals are positive. Trichotomy leaves no alternative.

(e) True: either the low- or high-degree coefficient ordering extends the given constant ordering.

(f) False: without $a > 0$, take $a = 0, b = 1$; no positive n has $1 < n0$.

(g) True: this is equivalent to the Archimedean definition. For $b > 0$ it is that definition; for $b \leq 0$, choose $n = 1$ since $a > 0 \geq b$.

(h) False: take $a = -1$, so $-a = 1$ is positive.

(i) False: at $a = 0$, neither a nor $-a$ is positive. One is positive when $a \neq 0$.

(j) True: $1, 2 \cdot 1, 3 \cdot 1, \ldots$ are distinct because a positive sum of 1's cannot be zero.

Exercise 25.16: Making pi larger than every rational

Since π is transcendental, evaluation $\phi_\pi : \mathbb{Q}[x] \to \mathbb{Q}[\pi]$ is an isomorphism. Give $\mathbb{Q}[x]$ the high-degree ordering and transfer it through this map. Equivalently a nonzero $a_0 + a_1\pi + \cdots + a_n\pi^n$ is positive exactly when its highest nonzero rational coefficient is positive. Then for every $q \in \mathbb{Q}$, $\pi - q$ has highest coefficient 1, so $\boxed{q < \pi}$ in this ordering. Transcendence ensures a unique polynomial representation, so the sign rule is well defined.

Exercise 25.17: Conjugation of a quadratic ring

Representations $m + n\sqrt{2}$ are unique by irrationality of $\sqrt{2}$. For $u = m + n\sqrt{2}$, $v = r + s\sqrt{2}$, direct calculation gives

$$\phi(u + v) = (m + r) - (n + s)\sqrt{2} = \phi(u) + \phi(v).$$

Since $uv = (mr + 2ns) + (ms + nr)\sqrt{2}$,

$$\phi(uv) = (mr + 2ns) - (ms + nr)\sqrt{2} = (m - n\sqrt{2})(r - s\sqrt{2}).$$

Thus multiplication is preserved as well. In fact ϕ^2 is the identity, so the homomorphism is an automorphism.

Exercise 25.18: Cone membership means greater than zero

By the definition in Theorem 25.5, $0 < a$ means exactly $a - 0 \in P$. This equals $a \in P$, which is the hypothesis. Thus the conclusion follows directly from the definition of $<$.

Exercise 25.19: Equal positive multiples

First, an ordered ring has no nonzero zero divisors: for nonzero u, v, choose signs so that $\pm u, \pm v \in P$; closure makes one of $uv, -uv$ positive, so $uv \neq 0$. Hence if $c = 0$, then $bd = 0$ with $b \neq 0$ forces $d = 0$, and similarly with $d = 0$.

Suppose now that neither is zero. If $c \in P$, then $ac \in P$. If $-d \in P$, closure would also give $b(-d) = -bd = -ac \in P$, contradicting trichotomy. Thus $d \in P$, and $cd \in P$. If instead $-c \in P$, the same argument applied to $a(-c) = b(-d)$ gives $-d \in P$, whence $cd = (-c)(-d) \in P$. This proves exactly the stated alternatives using only cone closure and trichotomy.

Exercise 25.20: Reversing signs reverses an inequality

The hypothesis $a < b$ means $b - a \in P$. But

$$(-a) - (-b) = b - a \in P.$$

By the definition of $<$, this is precisely $-b < -a$.

Exercise 25.21: A negative times a positive

From $a < 0$ we obtain $-a \in P$, and from $0 < b$ we obtain $b \in P$. Closure under multiplication gives $(-a)b = -ab \in P$. Since $0 - ab = -ab$ is positive, the defining condition for $<$ gives $ab < 0$. No separate sign rule has been assumed.

Exercise 25.22: A positive quotient

Since $b > 0$, it is nonzero and b^{-1} exists. The square of any nonzero element is in P: trichotomy makes either u or $-u$ positive, and squaring that element gives $u^2 \in P$. Thus $(b^{-1})^2 \in P$. Multiplying by $b \in P$ gives $b^{-1} = (b^{-1})^2 b \in P$. Finally closure gives $a/b = ab^{-1} \in P$, proving positivity of the quotient.

Exercise 25.23: Inverting a positive element below one

The assumptions give $a \in P$ and $1 - a \in P$. By Exercise 25.22, $(1 - a)/a \in P$. But this quotient is $a^{-1} - 1$. Therefore $1 < a^{-1}$ by the defining relation $<$.

Exercise 25.24: Inverting a negative element above minus one

Let $b = -a$. The inequalities $-1 < a < 0$ imply $0 < b < 1$, using the difference definition of $<$ or Exercise 25.20. Exercise 25.23 gives $1 < b^{-1}$. Reversing signs yields $-b^{-1} < -1$. Since $a^{-1} = (-b)^{-1} = -b^{-1}$, the conclusion is $a^{-1} < -1$.

Exercise 25.25: Transporting an ordering across an isomorphism

Let $P' = \phi[P]$. It is nonempty because P is nonempty. If $u = \phi(a)$ and $v = \phi(b)$ with $a, b \in P$, then $u + v = \phi(a + b) \in P'$ and $uv = \phi(ab) \in P'$ by closure of P. For any $u \in R'$, take its unique preimage a. Exactly one of $a \in P$, $a = 0$, $-a \in P$ holds, and injectivity and surjectivity translate this into exactly one of $u \in P'$, $u = 0$, $-u \in P'$. Hence P' is an ordering cone. Finally

$$\phi(a) <' \phi(b) \iff \phi(b) - \phi(a) = \phi(b - a) \in \phi[P] \iff b - a \in P \iff a < b.$$

The middle reverse implication uses injectivity of ϕ. This proves Theorem 25.10 in full.

Exercise 25.26: The induced ordering on a subring

For a nonzero subring S, put $P_S = P \cap S$. It is nonempty: choose $s \in S \setminus \{0\}$, and trichotomy gives either s or $-s$ in P_S. Sums and products of elements of P_S lie in P by cone closure and in S by the

subring axioms. For any $s \in S$, exactly one of $s \in P$, $s = 0$, $-s \in P$ holds; since both $s, -s$ lie in S, this is exactly the cone trichotomy for P_S. Thus the induced order is the restriction of the original order.

The nonzero qualification matters: for $S = \{0\}$ the intersection is empty, contrary to the definition's nonempty-cone requirement. If one also imposes the section's standing requirement that every ordered ring have nonzero unity, restrict the assertion to subrings with nonzero unity. For nonunital nonzero subrings the same proof still supplies the asserted order and cone axioms.

Exercise 25.27: Recovering the cone from the order relation

Set $P = \{x : 0 < x\}$. Since the ring is nonzero, compare 0 with some $a \neq 0$. Trichotomy gives either $0 < a$ or $a < 0$; in the latter case additive isotonicity gives $0 < -a$. Thus P is nonempty. If $a, b \in P$, adding a to $0 < b$ gives $a < a + b$, and transitivity with $0 < a$ gives $0 < a + b$. Multiplicative isotonicity applied to $0 < b$ and $0 < a$ gives $0 = a0 < ab$. So P is closed under both operations.

Trichotomy of 0 and x, and additive isotonicity translating $x < 0$ to $0 < -x$, give exactly one of $x \in P$, $x = 0$, $-x \in P$. Thus this is a positive cone. Finally additive isotonicity, in both directions, gives

$$a < b \iff 0 < b - a \iff b - a \in P.$$

Therefore the relation induced by P is the original relation, proving the converse in Theorem 25.5.

Exercise 25.28: Injectivity of an odd power in an ordered domain

An odd power has the same sign as its nonzero base: a positive base gives a positive product; for a negative base, its negative is positive and $(-c)^{2n+1} = -c^{2n+1}$. Thus equal odd powers cannot come from bases of opposite signs, or from one zero and one nonzero base.

For positive $a < b$, factor

$$b^{2n+1} - a^{2n+1} = (b - a)\big(b^{2n} + b^{2n-1}a + \cdots + a^{2n}\big).$$

Every summand in the parenthesis is positive, so the difference is positive and equality of the powers is impossible. The case $0 < a = b$ already has the conclusion. If $a, b < 0$, their negatives are positive, and equality of odd powers gives equality of the odd powers of $-a, -b$, reducing to the positive case. Trichotomy therefore leaves only $\boxed{a = b}$.

Exercise 25.29: Eight different orderings in two variables

For each stage choose low (L) or high (H) degree to determine the sign. Applying the univariate construction twice gives eight orderings. To distinguish them, record whether four fixed polynomials

are positive (+) or negative (−):

first ring, then outer variable	inner	outer	$x-1$	$y-1$	$x-y$	$xy-1$
$(R[x])[y]$	L	L	−	−	+	−
$(R[x])[y]$	L	H	−	+	−	+
$(R[x])[y]$	H	L	+	−	+	−
$(R[x])[y]$	H	H	+	+	−	+
$(R[y])[x]$	L	L	−	−	−	−
$(R[y])[x]$	L	H	+	−	+	+
$(R[y])[x]$	H	L	−	+	−	−
$(R[y])[x]$	H	H	+	+	+	+

.

For example, with $(R[x])[y]$ and outer L, the deciding y^0 coefficient of $x-y$ is $x>0$, while that of $xy-1$ is $-1<0$. With outer H, their deciding coefficients are -1 and x, reversing those two signs. The other rows follow by interchanging x, y and choosing the specified inner signs for $x-1$ or $y-1$. No two sign rows agree. Thus no two of the eight orderings can be the same.